

Infections and Intoxications from the Ocean: Risks of the Shore

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ABSTRACT Marine and coastal regions provide a diverse range of foods and serve as recreation and leisure areas for large numbers of people in many parts of the world. However, they also serve as environments associated with numerous hazards. The number of cases of fish- and shellfish-related food poisonings in the United States has increased in recent years, accounting for over 600,000 illnesses, with 3,000 hospitalizations and 94 deaths annually. Human diseases due to pathogenic *Vibrio* species can result from both ingestion of contaminated shellfish and exposure of open wounds to contaminated seawater. A variety of infections may result from human interactions with marine life, including sharks, barracudas, and moray eels. This chapter covers some of the risks of the shore, including fish and shellfish intoxications, infections related to *Vibrio* species, and infections resulting from marine trauma.

FISH AND SHELLFISH INTOXICATIONS

Introduction

In the United States, the consumption of seafood in 2012 alone totaled more than 4.7 billion pounds, or 14.4 pounds per person per year. An increase in the consumption of seafood is occurring with a resultant increase in the number of cases of fish- and shellfish-related food poisonings. Improved reporting of cases due to a greater awareness of the public and health care personnel of the association between seafood consumption and illness has also contributed to this observed increase. In the United States, food-borne illness due to contaminated fish and shellfish account for over 600,000 illnesses, including 3,000 hospitalizations and 94 deaths every year. Food-borne disease from fish and shellfish can be categorized into allergic, infectious, and toxin-mediated etiologies. The CDC and U.S.

Department of Health and Human Services reported on all types of food-borne illness in the United States from 1998 to 2008. Bacterial causes were involved in more than 77% of confirmed outbreaks, whereas chemical poisonings from fish and shellfish toxins were responsible for 11.4% of outbreaks of confirmed etiology.

Vertebrate fish intoxication can be divided into three groups: (i) ichthyosarcotoxic fish contain toxin in their viscera, mucous, skin, or musculature; (ii) ichthyotoxic fish contain toxin in their gonads; and (iii) ichthyohemotoxic fish contain toxin in their blood. At least nine types of ichthyosarcotoxism are known, of which ciguatera, scombroid, and pufferfish poisoning are the most common ([Table 1](#)).

The term “shellfish” includes crustaceans, which are mobile animals that have a hard articulated exoskeleton, and mollusks, which have hard shells and are sedentary or have limited locomotion. Crustacean species include lobsters, shrimp, crabs, scampi, and crawfish. Mollusks can be divided into the bivalves, which have two shells joined by a hinge, and the gastropods, which have a whorled snail-like shell. The bivalves include oysters, mussels, clams, and scallops. Gastropods of commercial importance include whelks and periwinkles. With the

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TABLE 1 Fish and shellfish poisoning

Disease	Source	Toxin	Mechanism	Epidemiology	Clinical data	Mortality	Treatment
Shellfish poisoning							
Paralytic shellfish ^a	Alaskan butter clam (<i>Saxidomus giganteus</i>), mussels, oysters, scallops, cockles	Saxitoxin (heat stable)	Like tetrodotoxin; blocks Na ⁺ channels	Low temp: >30° N or S; 10 outbreaks with 63 cases from 1971–1977 in New England, Southwest, Alaska	0.5–3 h; paresthesias or dysesthesias of mouth and extremities; 14% with N/V/D ^e	8–9% (usually within 12–24 h)	Supportive
Neurotoxic shellfish ^b	May be aerosolized in the surf	Brevetoxins (heat and acid stable) A, B, C (polycyclic ethers)	Probably via altering Na conductance	Gulf coasts of Florida and Texas	<3 h; paresthesias; temp reversal; cerebellar and GI ^f symptoms	Rare	Supportive; B ₂ agonists; cholinergic antagonists; Ca ²⁺ channel blockers
Amnesic shellfish (neurovisceral toxic syndrome)	<i>Pseudo-Nitzschia pungens</i> diatoms in mussels	Domoic acid (heat stable)	Neuroexcitatory like glutamic acid; causes hippocampal necrosis	Atlantic, Pacific, and Indian Oceans	0.25–38 h ($\bar{x}$ = 5.5 h); N/V, GI bleeding; HA ^h ; memory loss	3%	
Diarrhetic shellfish	Shellfish with dinoflagellate: <i>Dinophysis fortii</i> or <i>Dinophysis acuminata</i>	Okadaic acid	Blocks phosphatase that degrades A/G kinase products	Japan and The Netherlands	5–6 h (range, 0.5–30 h); N/V/D with abdominal cramps		
Fish poisoning (ichthyosarcotoxic)							
Puffer fish ^c	Puffer fish, shellfish, salamanders, frogs, newts	Tetrodotoxin (heat stable, nonprotein)	Blocks Na channels; axonal transmission	Japan, New Jersey, Long Island	10–45 min (<3 h); paresthesias; weakness; paralysis; hypotension; bradycardia; respiratory paralysis	59% (in first 24 h)	Supportive
Ciguatera ^d	Larger (>5 lb) carnivorous reef fish (barracuda, herring, jacks, grouper, snapper, moray eels, etc.)	Ciguateroxins and maitotoxin (heat-stable tasteless polyether)	Blocks C ²⁺ regulation of Na ⁺ channels (opens voltage-dependent Na ⁺ channels)	<35° N and S; subtropical/tropical (Florida, Hawaii, S. Pacific); most common marine food poisoning in United States	2–30 h 1 or 1 BP ^g ; N/V/D with abdominal pain; temp reversal; teeth "loose"	<12% (rare)	Symptomatic (mannitol; opiates are dangerous)
Scombroid fish	Tuna, mackerel, skipjack, bonito, albacore, bluefish, mahimahi	Histamine (heat stable) from bacterial decarboxylation of histidine	Histamine reaction	Common worldwide, especially coastal United States	Minutes–hours ($\bar{x}$ = 30 min); flush; HA, dizziness, burning	Rare	Antihistamine (intravenous cimetidine)

^aPrimary causes are red tides and dinoflagellates such as *Alexandrium catenella* and *Alexandrium tamarenis*.^bPrimary causes are red tides and unarmored dinoflagellates such as *Karenia brevis*.^cTetraodontidae; also known as *fugu*.^dThe primary cause is the dinoflagellate *Gambierdiscus toxicus*.^eN/V/D, nausea, vomiting, diarrhea.^fGI, gastrointestinal^gBP, blood pressure.^hHA, headache.

exception of scallops, which reside in deeper waters, all mollusks grow in and are harvested from nearshore coastal waters. Ingestion of shellfish containing toxins produced by dinoflagellates may induce dramatic and sometimes fatal illness.

Dinoflagellates and Red Tides

Dinoflagellates, or plankton, are unicellular plant-like organisms with a worldwide distribution which serve as an important element of the food chain of marine animals. During blooms, these organisms may achieve concentrations high enough to impart a reddish or yellow discoloration to the sea due to the local production of neurotoxins and pigmented proteins—hence the name “red tide.” There are 15 species of toxic dinoflagellates known to inhabit the waters surrounding the United States, of which four are related to human poisoning and six are associated with the formation of red tides.

The association of red tides with human illness has been known since ancient times, the earliest description being from the bible (Exodus 7:20-21): “And all the water that was in the Nile was turned to blood. And the fish that were in the Nile died, and the Nile became foul, so that the Egyptians could not drink water from the Nile.” North American Indians were aware of red tides and their association with poisoning due to mussel ingestion. In 1793, Vancouver described what may be the first description of poisoning due to shellfish ingestion among sailors exploring passages off the mainland coast of what is now British Columbia; several men became ill after eating roasted mussels, and one of them died within 5 hours. Walker, in 1884, described several people who became ill after eating oysters in Florida, possibly related to a red tide.

Red tides can be caused by nontoxic dinoflagellates, and shellfish may become poisonous even in the absence of a red tide. Vectors of shellfish poisoning are mainly filter-feeders that ingest large quantities of these dinoflagellates, many of which are toxigenic. The continuous filtration can result in the accumulation of large quantities of toxin within the digestive glands of the shellfish or in the case of the Alaskan butter clam, the siphon.

Paralytic Fish Poisoning

Paralytic shellfish poisoning (PSP) results from the ingestion of marine mollusks containing potent neurotoxins, the best known being saxitoxin, named after the Alaskan butter clam, *Saxidomus*. Several other toxins are known, each of which shares the ability to invoke a

variety of biological effects including occasionally severe and sometimes fatal impairment in sensory, cerebellar, and motor function.

During 1973 to 1987, state health departments reported 19 outbreaks of PSP (mean: eight people) to the CDC's Food-Borne Disease Outbreak Surveillance System, which accounted for 1.1% of all outbreaks of food-borne disease in the United States in 1972 to 1977 and 1.0% in 1978 to 1982. The implicated mollusks included mussels, oysters, clams, scallops, and cockles. Pufferfish caused 13 cases of PSP in Florida in 2002. Worldwide, it was estimated that more than 1,600 cases occurred in 1974 alone, with more than 300 deaths. In 1990, two outbreaks occurred in the United States: one involving six people after ingesting mussels harvested off the Nantucket coast in Massachusetts and another in Alaska involving four people with one fatality. In May and June of 2011, 21 cases of PSP were identified in southeastern Alaska following the ingestion of contaminated cockles. The incidence of PSP in Old Harbor and Kodiak, Alaska, has been estimated at 15 and 1.5 cases per 1,000 persons per year, respectively. The case fatality ratio is about 8 to 9%, usually secondary to respiratory failure. No deaths, however, were reported in 42 outbreaks involving 94 cases reported in Alaska between 1976 through 1989. In one analysis of two outbreaks of PSP in Alaska, those residents who knew nothing about PSP reported the same frequency of symptoms as those who knew about the potential lethal effects of PSP.

The effects of PSP poisoning are harmful not only to humans but also to fish, birds, and other wildlife that rely on aquatic sources of food. The ecologic consequences can be devastating. One of the earliest signs of a toxic bloom is the sudden and unexplained death of large numbers of fish and wildlife in the vicinity of a bloom. The American Indians were aware of these associations and avoided fish and shellfish ingestion during such times. These ecologic effects may persist for months following the onset of an outbreak and require a year or more for affected shellfish to become safe for human consumption. Economic consequences can be equally devastating because the shellfish industry becomes paralyzed during this period. Widespread reporting by the news media and strict adherence to public safety measures leads to the significant depression of demand for fish and shellfish not only in affected areas but in unaffected regional areas as well. It is estimated that the cost of surveillance and enforcement during outbreaks of PSP poisoning in the United States is about \$1.2 million per year.

The dinoflagellates responsible for paralytic fish poisoning are widely distributed globally, but outbreaks usually occur endemically in specific geographic areas. Such blooms are usually unpredictable and can occur with rapid accumulation of toxic concentrations. Most cases of PSP occur in cold, temperate waters above 30° N and below 30° S, but tropical cases have been reported in Thailand, Singapore, India, Guatemala, Malaysia, New Guinea, the Solomon Islands, Mexico, and El Salvador. Most North American cases occur along the Pacific coast from central California to Alaska and the Aleutian Islands and on the East coast in the New England coastal area as well as Nova Scotia, New Brunswick, and Quebec. The majority of outbreaks are reported in coastal areas, but inland cases have occurred, occasionally remote from the sea.

Along the West coast, cases tend to occur from May to October, and on the East coast, from July to September. The Alaskan butter clam can be dangerous year-round. The period of toxicity usually lasts for a few days during each outbreak, but toxic levels may persist for many months. Factors favoring toxic blooms include warm water temperatures (usually when water temperatures reach 16°C), periods of high solar radiation, optimal concentrations of trace vitamins and minerals, and periods of turbulence such as during hurricanes, dredging, and the transplantation of shellfish. Afflicted shellfish can be found along open coasts, in bays, and in estuarine areas. The most important hydrographic factor is probably a thermocline imposed barrier; i.e., areas where water temperatures are greater than 16°C. Occasionally, red tides may be precipitated by the lowering of salinity such as at sites of river discharge or during periods of heavy rainfall.

Many types of molluscan shellfish may become toxic. The 19 outbreaks reported to the CDC during 1973 to 1987 were due to the ingestion of mussels, clams, oysters, scallops, and cockles. In Alaska, during 1976 through 1989, 55% of the 42 reported outbreaks were due to the ingestion of the Alaskan butter clam. Other shellfish implicated in Alaskan outbreaks included mussels, cockles, steamer clams, sea snails, and razor clams. In addition to the ingestion of shellfish, cases of PSP have also been linked to the ingestion of mackerel, scads, and several species of crabs.

Mollusks become toxic when they ingest toxic dinoflagellates. The toxins accumulate in the digestive glands and can remain there for a long time. This toxic accumulation can occur even in the absence of the buildup of sufficient numbers of dinoflagellates to discolor the water. If placed in dinoflagellate-free seawater, it can

take up to 12 days for the shellfish to become nontoxic. In the case of the Alaskan butter clam, the toxins tend to accumulate in the siphon, from which they are eliminated very slowly. Scallops, on the other hand, do not accumulate toxins in the adductor muscle, which is the part usually ingested, although toxins may accumulate in the other tissues.

Several species of toxic dinoflagellates have been implicated in outbreaks of PSP. In North America, *Alexandrium catanella* is the principal dinoflagellate species responsible for outbreaks along the northwest Pacific coast of the continent, whereas *Alexandrium tamerensis* is responsible for most outbreaks along the north Atlantic coast in addition to some outbreaks on the Pacific coast. In the tropics, *Pyrodinium bahamense* is responsible for most outbreaks. Each of these species is armored, ranging in size from 25 to 46 microns, and has a tendency to become highly bioluminescent during blooms. During the winter months, most toxic dinoflagellates exist as cysts in the sediment beneath the sea. Turbulent conditions can disperse these cysts; when water temperatures reach optimal conditions (16°C), blooming occurs by excystment and the formation of motile cells.

There are 21 molecular forms of PSP toxins, of which saxitoxin is the primary toxin responsible for the biologic effects of PSP. Other toxins include neotoxin and the gonyautoxins, which are designated by roman numerals in the order of their discovery. These toxins are pharmacologically similar to tetrodotoxin and are estimated to be 50 times more potent than curare and 1,000 times more potent than cyanide. Saxitoxin is an alkaloid, nonprotein, low-molecular-weight, water-soluble toxin that is heat-stable. Cooking does not inactivate it, and it tends to become concentrated in broth. It bears some resemblance to guanine and may undergo trimethylation to form a toxic acetylcholine-like compound. The toxic effects result from binding of the toxin to either the cell membrane, a cation receptor, or both to inhibit sodium influx, thus blocking the action potential along neuronal axons or skeletal muscle. Higher concentrations may have a similar electrophysiologic effect on cardiac and smooth muscle. Evidence exists that under acidic conditions, as in the stomach or in pickling containers, nontoxic products may become converted to toxins. Furthermore, shellfish may be capable of converting nontoxins to potent neurotoxins. Children may be more susceptible to the toxic effects. The lethal dose of saxitoxin has been estimated to be 0.3 to 1.0 mg; a single mussel may contain 30 to 50 mg. The safe level of toxin in shellfish has been defined as 80 µg/100 g of shellfish.

The symptoms of PSP usually begin within 30 minutes after the ingestion of toxic shellfish but can occur up to 3 hours later. The incubation period appears to be inversely related to the amount of toxin ingested. Initially there may be paresthesias and dysesthesias of the lips, tongue, and face, which subsequently can progress to involve the neck, arms, fingertips, legs, and toes. Gastrointestinal symptoms, usually nausea, vomiting, and diarrhea, were seen in only 14% of patients in one series. Many people have described a “feeling of floating.” Progression of symptoms is usually dependent on the dose of toxin ingested; more severe cases may be accompanied by weakness, ataxia, incoordination, and cranial nerve findings such as bulbar paresis, iridoplegia, dysphonia, and dysphagia. Other associated symptoms include headache, salivation, intense thirst, and temporary blindness. High toxin intake can result in muscular paralysis and respiratory failure, which is the usual cause of death in fatal cases. The illness can be sufficiently severe to require hospitalization in 30 to 60% of affected patients. When it does occur, death usually ensues within 12 hours. If the patient survives 12 to 18 hours, the prognosis for recovery is good. Recovery usually occurs within a week without complications, but several patients in one outbreak described persistent headaches, memory loss, and fatigue lasting for several weeks.

The diagnosis is based on a history of recent shellfish ingestion in the appropriate clinical setting. Routine laboratory tests are usually nonspecific and not helpful in establishing the diagnosis. The CK-MB isoform of creatine kinase may be elevated in the absence of myocardial damage. Diagnosis can be confirmed by a standard mouse bioassay method in which toxin concentrations of the suspect shellfish are calculated by determining the dilution of a shellfish homogenate required to kill a 20 g mouse in 5 to 7 minutes. The value can then be used to calculate an absolute concentration by comparison with a known control. Drawbacks include a precision of $\pm 20\%$, interference from sodium chloride, and the need to keep a constant supply of mice. Other techniques have been developed such as a fluorimetric assay, an immunologic assay, a colorimetric assay, and high pressure liquid chromatography. Efforts to isolate the toxin from gastric contents have been limited.

Treatment is largely supportive. Attempts should be made to remove unabsorbed toxin through gastric lavage or the administration of a cathartic or enema. Mechanical ventilation may become necessary in the event of respiratory failure. Hemodialysis was used successfully in one patient. Atropine should be avoided because saxitoxin and its derivatives may be anticholinergic.

Neurotoxic Shellfish Poisoning

Karenia brevis (formerly *Gymnodinium breve* and *Ptychodiscus brevis*) is the dinoflagellate responsible for neurotoxic shellfish poisoning, a syndrome similar to PSP, but symptoms are usually milder and paralysis and respiratory failure do not occur. This dinoflagellate produces neurotoxins and is responsible for the formation of red tides off the Gulf coasts of Florida and Texas; occasionally, sea currents can carry the organism to Florida's Atlantic coast. In 1987, a red tide due to *K. brevis* formed off the coast of North Carolina and was associated with an outbreak of neurotoxic shellfish poisoning involving 48 people. The source of this bloom was probably a red tide carried from Florida's southwest coast by Gulf Stream currents. Five cases were reported to the CDC from 1970 through 1974. In 2013, a *K. brevis* bloom had a significant detrimental effect on tourism in southwest Florida after a massive bloom induced respiratory distress among beachgoers as well as a massive fish kill which washed ashore onto the beaches.

K. brevis ranges in size from 20 to 40 μm in length and about 13 μm in width. It has been referred to as a “naked organism” because it lacks the shell of polysaccharide plates which characterizes *Protogonyaulax* species and other dinoflagellates. The neurotoxins produced by the dinoflagellate probably don't accumulate in fish but concentrate in filter-feeding shellfish in the vicinity of a bloom. Shellfish are not affected by the toxins, but humans become affected by ingestion of shellfish containing high levels of neurotoxin. Shellfish which may become toxic include oysters, clams, coquinas, and other bivalve mollusca.

Five or more separate nonprotein toxins or toxin components are produced by *K. brevis*. These are termed “brevetoxins” and include brevetoxins A, B, C, and Gb-4. Brevetoxins A, B, and C are polycyclic ethers which have similar structures and may be convertible among each other. The toxins are lipid soluble, acid stable, and base labile; some are heat stable. They probably act by altering sodium conductance at or near sodium channels. Animal studies have demonstrated various and diverse biologic effects including smooth muscle contraction through postganglionic parasympathetic acetylcholine release in dogs, inhibition of neuromuscular transmission in skeletal muscle, central nervous system stimulation with cardiovascular and respiratory impairment in dogs and cats, and norepinephrine release from nerve endings in rats. Brevetoxin B has been used as a model for a red tide pigment toxin.

Symptoms of neurotoxic shellfish poisoning usually begin within 3 hours after ingestion of contaminated

shellfish and include circumoral paresthesias which progress to involve the pharynx, trunk, and extremities. Reversal of hot and cold temperature sensation as in ciguatera fish poisoning can occur, as well as cerebellar symptoms such as vertigo, ataxia, and incoordination. Gastrointestinal symptoms are common and include nausea, diarrhea, and abdominal and rectal pain. Bradycardia, headache and cramping of the lower extremities, and dilated pupils have been reported. Severe cases may be accompanied by convulsions and subsequent need for respiratory support. Paralysis and respiratory failure are not seen. No deaths are known to have been reported.

The diagnosis is based on clinical grounds in the appropriate setting. There is no known antidote, and treatment is supportive and symptomatic, including airway management, intravenous fluids, atropine, and pressors if required. Symptoms are self-limiting and usually resolve completely without sequelae in a few days. One patient in North Carolina required admission to an intensive care unit for severe symptoms of bilateral carpopedal tremor and myalgia, total body paresthesia, ataxia, and vertigo after the ingestion of 45 oysters. Recovery was complete in approximately 9 hours.

The lack of an armored shell allows for aerosolization of the dinoflagellate during red tides by turbulent surf. This results in a unique syndrome seen along Florida's beaches characterized by respiratory and conjunctival irritation with the development of a nonproductive cough, shortness of breath, lacrimation, rhinorrhea, and sneezing. Asthmatics may develop a wheeze. The syndrome, aerosolized red tide respiratory irritation, is reversible upon leaving the beach. Beta-2 agonists, cholinergic antagonists, and calcium channel blockers may alleviate some of the respiratory symptoms.

Amnesic Shellfish Poisoning

In 1987, a previously unrecognized illness occurred among several hundred people who had ingested mussels harvested from cultivation beds located in three river estuaries on the eastern coast of Prince Edward Island, Canada. The affected individuals developed an acute illness characterized by severe nausea, gastrointestinal bleeding, and a severe and protracted neurologic disorder which included disorientation, confusion, dizziness, seizures, coma, and a persistent memory loss. Many individuals required prolonged hospitalization, and several deaths were reported. This syndrome, known as neurovisceral toxic syndrome, was subsequently linked to domoic acid, a toxin produced by the diatom *Pseudo-nitzschia pungens* f. multiseries. A nontoxic form of

the species, *P. pungens* f. *pungens*, is a common coastal diatom during the warmer months in the estuaries of Prince Edward Island and Galveston, Texas, and is replaced by *P. pungens* f. *multiseries* when fall and winter storms occur.

The implicated organism is widely distributed in the coastal waters of the Atlantic, Pacific, and Indian Oceans, but not all strains have been shown to produce domoic acid. Gulf Coast oysters were recently shown to contain this toxin, indicating a potential for human poisoning in this area. Mussels from the Canadian outbreak were shown to have high concentrations of this toxin in their digestive tracts. During the outbreak, a substantial bloom of *P. pungens* was noted by marine biologists patrolling the area. A possible reason for this is that freshwater runoff from record-breaking storms that year may have stratified the ocean layers of the estuaries, thereby enhancing the nutrients at just the right time in the diatom's life cycle. Similar blooms have been recorded since the initial outbreak, but proper surveillance measures enacted for domoic acid after the initial outbreak have effectively protected the public as well as the shellfish industry. Shellfish can clear the toxin from their tissues once exposed to clean seawater free of the toxic diatom.

Domoic acid is a heat-stable neuroexcitatory amine similar to glutamic and kainic acids. Its effects are about 2 to 3 times more potent than kainic acid and 30 to 100 times more potent than glutamic acid. Extracts of seaweed containing this toxin have been used in Japan as an ascaricidal agent for many years, although concentrations are significantly less than those causing illness, and no adverse effects have been documented. In rats, domoic acid stimulates kainic acid receptors in the hippocampus and can produce limbic seizures, memory and gait abnormalities, and degeneration of the hippocampus. Autopsy reports of victims of the Canadian outbreak revealed neuronal necrosis and astrocytosis in several areas of the brain, particularly the hippocampus and amygdala.

At least 107 individuals were involved in the 1987 outbreak. Symptoms began 15 minutes to 38 hours (median, 5.5 hours) after the ingestion of mussels. In approximate order of appearance, they included nausea and vomiting (76% of the patients), abdominal cramps (50%), gastric bleeding, diarrhea (42%), incapacitating headache (43%), dizziness, confusion, loss of short-term memory (25%), weakness, lethargy, somnolence, coma, and seizures. Hospitalization was necessary for 18% of those affected, and death occurred in 3. The memory loss was anterograde in all but the most severe cases and

persisted for at least two years in a few individuals. Cognitive functioning remained intact. Cardiovascular instability was also noted, presumably due to the early excitatory effects of domoic acid. Alternating hemiparesis and ophthalmoplegia were noted in two patients. Evidence suggests that elderly individuals may be more susceptible to the neurologic effects of domoic acid, possibly due to diminished renal function.

Puffer Fish Poisoning

Ingestion of fish in the order *Tetraodontoidea*, which includes puffer fish, porcupine fish, and the ocean sunfish can result in an acute illness referred to as puffer fish poisoning or tetraodotoxiation. This syndrome, although rare in the United States, is not uncommon in Japan, where 6,386 cases were reported in a 78-year period, with a mortality rate of 59% (Fig. 1). Known as *fugu-fugu* in Japan, puffer fish there is considered a delicacy and is specially prepared by trained individuals who require licenses to serve this popular dish, which can cost up to \$400 for one meal. In spite of these precautions, about 50 cases of *fugu* poisoning occur annually in association with these specialty restaurants. Although personal importation of *fugu* into the United States is illegal, the FDA has permitted it to be imported and served in Japanese restaurants in the United States by certified *fugu* chefs on special occasions. Captain Cook, on his second voyage, ate a piece of puffer fish liver and became acutely ill, requiring four days to

recover. Species of fish in this family are widespread and found in warm and temperate waters throughout the world; some species are eaten in New Jersey and Long Island. In April 1996, three cases of *fugu* poisoning occurred in three chefs in San Diego who shared pre-packaged, ready-to-eat *fugu* illegally imported from Japan.

Tetrodotoxin is a heat-stable nonprotein toxin which is concentrated in the liver, ovaries, and intestine of infected fish. Tetrodotoxin is now known to be produced by certain symbiotic bacteria that reside in these animals, such as *Pseudoaltermonas tetraodonis* and certain species of *Pseudomonas* and *Vibrio*. It is not unique to puffer fish and has been isolated from six classes of animals including shellfish, the blue-ringed octopus, salamanders, and a newt found on the campus of Stanford University. Certain frogs and newts in Central and South America harbor this toxin, and their skins are used to manufacture poison darts.

Tetrodotoxin has its own receptor located at or near the sodium channel of the external nerve axon cell membrane. Binding at this receptor inhibits nerve impulse propagation along preganglionic cholinergic, somatic motor, sensory, and sympathetic nerves in the central, peripheral, and autonomic nervous systems. Direct effects on the brainstem medulla can induce emesis or hyperemesis and respiratory depression. It is not a curare-like agent and does not act directly on the acetylcholine receptor at the motor endplate. Structurally, it resembles morphine, which may explain some of its narcotic activity. Primary pharmacologic effects include local anesthesia, hypotension, hypothermia, emesis, respiratory depression, and decreased systemic vascular resistance. High levels of toxin can inhibit impulse relay in skeletal or cardiac muscle.

Symptoms normally begin within 3 hours of ingestion of affected fish (usually 10 to 45 minutes). Symptoms include lethargy, weakness, paresthesias (including a numbness of the face and extremities), a floating sensation, emesis or hyperemesis, ataxia salivation, and dysphagia. The extent of symptomatology varies with the amount of toxin ingested. Muscular weakness can progress to total paralysis, including respiratory paralysis. Hypotension, bradycardia, and fixed dilated pupils can occur in severe cases. Symptoms usually resolve over a period of days; the prognosis is good if the patient survives 18 to 24 hours. Death occurs in about 60% within the first 24 hours.

The diagnosis is based on clinical grounds. Treatment consists of airway support and volume expansion with intravascular fluids such as normal saline and possibly

FIGURE 1 The puffer fish is considered a delicacy in Japan and must be prepared by specially trained chefs to avoid *fugu* poisoning. Photograph © Laszlo Ilyes 2006. Licensed under CC-BY-2.0 (<https://creativecommons.org/licenses/by/2.0/deed.en>). doi:10.1128/microbiolspec.IOL5-0008-2015.f1



pressors. Attempts should be made to remove unabsorbed toxins through gastric lavage and emesis. Some evidence exists that gastric lavage with 2% sodium bicarbonate is effective if used within the first hour of intoxication. Atropine is useful in the management of bradycardia. The anticholinesterases edrophonium, physostigmine, neostigmine, and galanthamine may have some benefits, as may veratrine-like agents and cysteine, although none of these have been proven in large-scale studies. Apomorphine currently appears to have the best antiemetic properties. Other potential treatment options include hyperbaric oxygen, the narcotic antagonist naloxone, and possibly monoclonal neutralizing antibodies.

Ciguatera Food Poisoning

Ciguatera is a distinct clinical syndrome that may follow the ingestion of certain tropical reef fishes which have acquired toxicity through the food chain. The name was given by Don Antonio Parra in Cuba in 1787 from the Spanish *cigua*, which refers to the poisonous turban shellfish *Turbinidae*. Sailors with Captain Cook suffered from this malady during his voyages. Outbreaks of ciguatera occur in tropical and subtropical regions between 35°N and 35°S latitudes. More than 50,000 cases occur globally every year. Ciguatera is the most commonly reported food-borne illness of marine origin in the United States; overall, it accounted for 5.8% of all disease-related food-borne outbreaks from 1972 to 1977 and 7.4% of outbreaks from 1978 to 1982. In endemic areas of the United States, 5 to 70 cases per 10,000 people are reported annually, with 90% of cases occurring in Florida and Hawaii. In 1987, an outbreak occurred in North Carolina involving 10 people in what is probably the first case of ciguatera associated with consumption of fish harvested from mainland U.S. coastal waters outside of Florida. In 1995, an outbreak of ciguatera occurred in U.S. soldiers serving in Haiti who had eaten locally caught fish. In New York City, 6 outbreaks of ciguatera fish poisoning linked to barracuda and grouper purchased from a local fish market in Queens involved 28 people between August 2010 and July 2011. A number of cases have been reported in nontropical areas such as Vermont and Iowa due to the retail of affected fish from Florida. In some areas of the world, the disease is so prevalent that only the largest outbreaks are reported. An epidemiologic analysis reported on more than 3,000 cases in the South Pacific. On one Pacific atoll, a 43% annual incidence was found during a routine survey of households in one such epidemic.

Gambierdiscus toxicus is the primary dinoflagellate responsible for the production of a number of closely related but distinct toxins responsible for the complex symptomatology of ciguatera. Other dinoflagellates may also be toxigenic including *Procentrum lima*. These dinoflagellates adhere to dead coral, bottom-associated marine algae, and seaweed. Herbivorous fish ingest the dinoflagellates, and the toxins are subsequently passed up the food chain to the larger carnivorous reef fish that concentrate the toxins in their tissues and result in human poisoning. Toxins are ultimately accumulated in viscera, although muscle tissue may also contain lethal amounts. Contamination of reef fish is more likely to occur during storms or other periods of turbulence. Larger fish are more likely to be contaminated. Fish are unaffected by the toxins. More than 400 species of fish have been implicated in ciguatera including anchovy, barracuda, filefish, herring, jacks, moray eels, oceanic bonito, parrot fish, porgie, seabass or grouper, red snapper, squirrelfish, surgeonfish, triggerfish, trunkfish, and wrasse. In Miami, the high incidence of ciguatera due to the ingestion of barracuda has resulted in the ban of the sale of this fish.

The toxins recovered from fish implicated in ciguatera include ciguatoxin(s), maitotoxin, lysophosphatidylcholine (maitotoxin-associated hemolysin), scaritoxin(s), palytoxin, and ciguatoxin-associated ATPase inhibitor. The toxins are lipid soluble, colorless, odorless, and heat stable; thus, affected fish lack any unusual taste, odor, or appearance, and cooking does not inactivate the toxins. Ciguatoxin is a polyether compound that probably acts by competitive inhibition of calcium regulation of the sodium channel. For this reason, calcium gluconate has been advocated in the treatment of ciguatera, although its efficacy has not been proven. The toxin appears to act by opening voltage-dependent sodium channels in all cell membranes with initial neural stimulation followed by conduction block primarily in skeletal muscle and neuronal membranes and less so in cardiac muscle. Higher doses can result in phrenic nerve paralysis and respiratory arrest. Maitotoxin is cardiotoxic and is also the most potent marine toxin known. Its cardiotoxic effects are due to enhanced calcium influx through the cardiac membrane with a resultant calcium overloaded state. These effects are abolished by verapamil in a rat model. In ciguatera, maitotoxin can result in hypotension, whereas ciguatoxin has hypertensive effects; the former may be responsible for ciguatera shock, whereas the latter may play a role in the chronic hypertension seen in chronic ciguatera. Immune sensitization to polycyclic ethers occurs in ciguatera by a T-cell-dependent

mechanism which results in serotonin release via abnormally released IgE. Subsequently, hypotension can ensue in response to certain medications (e.g., paraldehyde), foods, and factors generated in shock (e.g., thromboxane A₂). Morphine may form polycyclic ethers on epoxidation or endoperoxidation of the olefin moiety; this may account for the dramatic hypotension seen in some victims of ciguatera when administered opiates.

The symptoms of ciguatera have an ethnic variation which may be due to differences in diet; people of Philippine or Chinese extraction are more severely affected, Hawaiians least affected, and other groups interposed between. Overall, 175 symptoms have been noted. The incubation period has been reported to be from 2 to 30 hours. Gastrointestinal symptoms such as watery diarrhea, nausea and vomiting, and abdominal pain tend to predominate early, followed by neurologic symptoms, although great variation exists. Virtually all experience gastrointestinal symptoms during the course of the illness, which usually resolve in 24 to 48 hours. Myalgia and weakness, particularly of the lower extremities, may occur at any time. Intense generalized pruritus may occur, and women may even complain of pruritus of the vaginal vault. Painful intercourse has been described in afflicted individuals as well as their sexual partners. Bradycardia with hypotension occurs in 10 to 15% of cases; higher doses of toxin(s) may elicit a biphasic response where the bradycardia and hypotension are followed by tachycardia and hypertension. Shock and respiratory failure may occur within minutes. Skin lesions and a distinct erythematous desquamative rash have occasionally been reported. Fever, lacrimation, severe muscle spasms, and dysuria may also occur.

Initial neurologic symptoms usually consist of circumoral and distal paresthesias. Vertigo and ataxia are common and are accompanied by a wide variety of other neurologic manifestations such as cranial nerve palsies, motor paralysis, blurred vision, and coma. Pregnant women may complain of bizarre seizure-like fetal activity. Temperature reversal is an unusual characteristic commonly seen in ciguatera which usually occurs in 2 to 5 days. Hot objects may seem cold and cold objects can elicit an electric shock-like sensation. Serious thermal injury has been reported due to the individual's failure to recognize extreme heat as being such. In the Bahamas, natives may be seen holding beer cans wrapped in towels to keep their fingers from touching the cold metal. Teeth may seem painful or loose. Nightmares are quite common, whereas other individuals may complain of auditory hallucinations and zoopsia. All food may taste metallic.

Symptoms may wax and wane during any 24 hour period and give a "pseudodiurnal periodicity." Alcohol may exacerbate or induce the recurrence of symptoms in up to 28% of recovered patients. Other conditions that increase blood flow such as increased temperature or physical exertion can also exacerbate symptomatology. Whereas gastrointestinal symptoms usually resolve in 24 to 48 hours, neurologic, musculoskeletal, and cardiovascular symptoms may persist for months to years. Furthermore, recurrence of symptoms may intermittently appear for several months after recovery. Total duration of symptoms is usually several days to months, with neurologic symptoms and pruritus being the slowest to resolve. Sensitization is common, and immunity does not occur. Some individuals may never eat fish again because exposure to even minuscule amounts of ciguatera toxins may reproduce dramatic symptoms. Malignancy has developed in a few individuals. Death is rare and occurs in up to 12% of reported cases. No deaths occurred in 184 cases reported to the CDC in 1970 to 1974.

The diagnosis is based on clinical grounds. Laboratory abnormalities, if present, are usually due to fluid and electrolyte disturbances secondary to gastrointestinal manifestations. Elevated serum ammonia levels with abnormal prothrombin and partial thromboplastin times may reflect liver toxicity. When severe muscle spasms are present, there may be marked elevations of creatinine phosphokinase, serum glutamic oxaloacetic acid, and lactic acid dehydrogenase. Reversible T-wave changes on electrocardiograms have been reported. The toxins may be detected by a mouse bioassay which is subject to the same limitations as detection of saxitoxin and its derivatives in PSP (see above). Radioimmunoassay, often referred to as a "poke" or "stick" test, and chromatography have also been utilized to detect ciguatera toxins, though they are expensive and subject to other limitations as well.

Treatment is primarily symptomatic and supportive. Attempts to remove toxin through emesis or gastric lavage with activated charcoal should be made if vomiting has not occurred. A cathartic may be administered to remove toxin from the lower intestinal tract. Intravenous mannitol provided rapid and dramatic relief in 24 patients with ciguatera in one study, with rapid recovery from shock and coma in 2 patients. The effect of mannitol on the course of ciguatera is unknown but may be due to competitive inhibition of the toxin(s) on the sodium channels or the neutralization of toxin(s). More recent double-blind randomized trials have failed to demonstrate any benefit of intravenous mannitol over

normal saline, so the use of mannitol is controversial at present. Atropine may be used to control symptomatic bradycardia. Calcium gluconate and dopamine infusions have been effective in the treatment of hypotension. Amitriptyline and gabapentin have been used successfully to alleviate the paresthesias. Nifedipine has been shown to be of some benefit in alleviating headaches, but it should be used with caution in the early stages to avoid hypotension. Pruritus has responded to antihistamines and avoidance of alcohol, excessive exercise, and high ambient temperatures. Myalgia may respond to acetaminophen and indomethacin. Opiates and barbiturates should be avoided because they may aggravate hypotension.

A ciguatera diet has been devised to be used in the treatment of affected patients. It consists of a diet high in protein, carbohydrates, and vitamins and the avoidance of fish, shellfish, seeds, nuts, mayonnaise, and their products. In addition alcohol, marijuana, solvents, herbicides, insecticides, glues, epoxies, ethers, resins, and cosmetics should be avoided as well. These restrictions should be maintained for at least 3 to 6 months after complete resolution of symptoms, and probably 12 months in the severely affected. Ingestion of fish weighing more than 2.3 kg (5 pounds) or fish caught during red tides should be avoided by the general public because these are more likely to contain ciguatera toxins. To avoid dyspareunia, sexual activity should be avoided until symptoms abate, and afflicted males should at least consider the use of a condom if avoiding abstinence to prevent the transmission of dyspareunia to their unaffected partners.

Scombroid-Fish Poisoning

Scombroid-fish poisoning is an acute clinical syndrome characterized by symptoms of histamine toxicity resulting from the ingestion of spoiled fish. It represents the most common form of ichthyosarcotoxism in the world. In the United States, scombroid-fish poisoning accounts for less than 0.5% of food-borne illnesses and 40% of seafood-borne illness outbreaks. Between 1998 and 2002 167 scombroid outbreaks that affected 703 people were reported to the CDC. Thirty-eight individuals were hospitalized and no fatalities were reported. In March 1998, 24 people on a Hollywood movie set became ill after eating trays of escobar, a trendy seafood that has caught on with many West Coast crowds. The escobar was traced to a Florida supplier who had also sent a shipment to a celebrity ski resort in Utah, where several other cases were reported. Tuna burgers were the cause of 5 outbreaks of scombroid-fish poisoning involving

18 people in North Carolina from July 1998 to February 1999. Most cases in the United States are reported from coastal states and Hawaii, although cases have occurred in the Midwest.

The disease is associated with ingestion of fish which belong to the families *Scombroidea* and *Scomberesocidae*, which includes tuna, mackerel, skipjack, bonito, and albacore. Nonscombroid fish, such as mahi-mahi, bluefish, amberjack, herring, sardines, and anchovies, as well as cheese, have also been implicated. Scombroid fish are distributed worldwide throughout temperate and tropical waters and have occasionally been found in polar waters.

Histamine has been identified as the toxin responsible for the symptoms of scombroid-fish poisoning. The affected fish do not contain high levels of histamine in their flesh at the time of capture; instead, they contain histidine. Histamine is produced during the process of spoilage by the enzymatic decarboxylation of histidine by certain marine bacteria, particularly *Morganella morganii*, *Klebsiella pneumoniae*, *Escherichia coli*, clostridia, *Achromobacter histamineum*, *Plesiomonas shigelloides*, *Enterobacter intermedium*, *Serratia marcescens*, *Serratia plymuthica*, *Serratia fonticola*, and *Hafnia alvei*, which are common surface bacteria on fish. This occurs optimally at temperatures between 20°C and 30°C. Typically, this takes place when previously refrigerated fish is allowed to warm for a period of time before it is prepared, and for this reason, the most effective means of preventing scombroid-fish poisoning is proper refrigeration to temperatures $\leq 40^{\circ}\text{F}$ ($\leq 4.4^{\circ}\text{C}$) at all times between catching and consumption. Histamine is heat stable and not destroyed by cooking. It is also stable in freezing temperatures. Oral histamine administered in large doses is rapidly metabolized in the liver and intestinal mucosa and symptoms, if present, are generally mild. For this reason the presence of an unknown synergistic substance(s) has been proposed to account for the high levels of histamine noted in individuals afflicted with scombroid-fish poisoning.

Symptoms generally appear within several minutes to several hours of ingestion (median of 30 minutes), with a median duration of 4 hours. Symptoms may persist for 12 to 24 hours. The fish has been occasionally described as being sharp, peppery, or bitter but usually not as being unpleasant. Symptoms initially appear as flushing and a hot sensation of the skin, dizziness, headache, a burning sensation in the mouth and throat, shortness of breath in the absence of bronchospasm, itching with or without urticaria, and palpitations. Gastrointestinal symptoms appear as diarrhea, nausea, and rarely vomiting. A sunburned-appearing skin rash with sharply

demarcated borders may develop as well as conjunctival injection. More severe symptoms include difficulty swallowing, respiratory distress with bronchospasm, hypotension, tachycardia, and blurred vision. People receiving isoniazid and other inhibitors of endogenous histaminase may have more severe symptoms. Deaths have occurred but are very rare.

The diagnosis is made on clinical grounds and is usually fairly evident because the incubation period is relatively short and several people are usually affected at once. Many people are erroneously diagnosed as having a “fish allergy” and are told to abstain from eating fish for the rest of their lives. Laboratory data is usually not helpful. Levels of histamine and its metabolite *N*-methylhistamine are elevated in urine samples at the time of onset of symptoms and may persist for more than 24 hours, although this is not routinely tested. Laboratory confirmation of scombroid-fish poisoning is accomplished by measurement of histamine in the suspect fish. The concentration may vary from one portion of the fish to another, so several areas must be sampled. The FDA has established the maximum safe level of histamine in tuna to be 450 μmol per 100 g of fresh tuna; fresh tuna contains levels less than 9 μmol per 100 g. In one recent outbreak in Tennessee, marlin was the implicated fish and contained levels of histamine greater than 2,500 μmol per 100 g.

Treatment is supportive and symptomatic. If the gastrointestinal symptoms are not severe, gastric lavage or catharsis may be employed to remove unabsorbed histamine. Depending on the severity of symptoms, management can best be accomplished by the use of any one or a combination of agents such as epinephrine, oxygen, diphenhydramine, hydroxyzine, and corticosteroids. Aminophylline may be used for the rare case of severe respiratory distress due to bronchospasm. Intravenous cimetidine has been reported to provide rapid and complete resolution of symptoms in severe cases that have not responded to antihistamines. Caution must be exercised in the simultaneous use of H1 and H2 blockers to avoid hypotension.

Diarrheic Shellfish Poisoning

This clinical syndrome has not been reported in the United States to date but has been implicated in a number of short-lived outbreaks of acute onset of diarrhea following shellfish ingestion in other parts of the world, particularly Japan and The Netherlands. It is caused by the ingestion of okadaic acid and other toxins concentrated in shellfish that feed on the dinoflagellates that produce the toxins. Okadaic acid does not directly

stimulate intestinal secretion, but instead causes a significant increase in paracellular permeability. From 1976 to 1982, more than 1,300 cases were diagnosed in Japan, while sporadic cases occurred in the Netherlands and Chile. In 1989, 150 people on the Adriatic Coast of Italy were afflicted by this illness after the ingestion of contaminated mussels. This was the first case of diarrheic mussel poisoning observed in the Mediterranean area. Okadaic acid has been detected in Gulf of Mexico shellfish and phytoplankton.

Dinophysis fortii is the responsible toxin-producing dinoflagellate in Japan, whereas *Dinophysis acuminata* produces the toxin in outbreaks occurring in The Netherlands. Mussels, clams, and scallops are the implicated shellfish that cause human outbreaks. Outbreaks are associated with dinoflagellate blooms.

Symptoms usually begin about 5 to 6 hours after shellfish ingestion, with a range of 30 minutes to 12 hours. Although usually mild, the severity of the illness is dependent on the amount of toxin ingested. The symptoms consist of diarrhea, abdominal cramps, nausea, and vomiting. No fatalities have been reported, and recovery generally occurs within 2 days. Treatment is supportive because the symptoms are self-limited.

Pfiesteria piscicida

P. piscicida is an estuarine dinoflagellate first described in 1991. This usually nontoxic organism feeds on aquatic organic material but can produce toxins that can kill fish. The toxin can induce formation of open ulcerative lesions, hemorrhaging, and death of fish and shellfish. Beginning in autumn 1996, fish with “punched-out” skin lesions and erratic behavior caused by exposure to toxins produced by *P. piscicida* or *Pfiesteria*-like species were seen in the Pocomoke River and adjacent waterways on the eastern shore of Maryland. In August 1997, similar fish kills were again reported. That same month, 24 sportsman, environmental workers, and commercial fisherman who had contact with the water reported illness. Human illness, known as possible estuary-associated syndrome, has been a topic of much debate due to the lack of specific testing and possible implication of unrelated factors. Possible estuary-associated syndrome is not an infectious disease, and there have been no cases associated with eating fish or shellfish harvested from waters where *P. piscicida* has been found. Prior to 1998, 37 cases were reported to the CDC, and very few since then. The reason for the drop-off in cases since 1998 is likely due to the paucity of “fish events” due to *P. piscicida* since June 1, 1998.

Symptoms were more likely to occur in those with the highest exposure to the contaminated water. These include neuropsychiatric symptoms (including new or increased forgetfulness), severe respiratory distress including asthma, headache, narcosis, severe stomach cramping, nausea with vomiting, and eye irritation with reddening and blurred vision. Neuropsychiatric symptoms tend to last the longest and completely resolve by 6 months, usually by 10 to 12 weeks.

The reason why *Pfiesteria* suddenly became toxic is unknown, but it has been linked to coastal chicken and hog farmers in Maryland and North Carolina.

THE VIBRIOS

Introduction

Prior to the 1960s, the focus on vibrios as pathogens of human disease was on *Vibrio cholerae*, the etiologic agent of cholera. Pacini originally described a vibrio-like organism as the etiologic agent of Asiatic cholera in 1854. The organism was not isolated until 32 years later by Koch, who called the bacillus *Kommabacillus*, referring to its curved shape. *V. cholerae* was the only currently recognized vibrio known to cause human disease until 1951, when Fujino described a bacterium resembling *V. cholerae* which was responsible for an epidemic of acute gastroenteritis in Japan involving 272 people and 20 deaths. He named this organism *Pasteurella parahaemolyticus*, which was subsequently placed in the genus *Vibrio* in 1963. Two distinct biotypes were recognized at that time which were found to be separate species. In 1968, biotype 1 became known as *Vibrio parahaemolyticus*, whereas biotype 2 became known as *Vibrio alginolyticus*. Since then, more than 80 species of *Vibrio* have been recognized, of which at least 12 are known to be pathogenic to humans. Furthermore, several unnamed species have been recently identified, so the list is likely to grow.

Pathogenic members of the genus *Vibrio* are Gram-negative, curved, rod-shaped facultative anaerobes which are capable of both fermentative and respiratory metabolism. They are motile organisms which measure 1.5 to 3.0 μm in length and 0.5 to 0.8 μm in width (Fig. 2). They contain a single, sheathed polar flagellum in liquid medium and occasionally may display shorter lateral flagella on solid media. They are anaerogenic (with the exception of *Vibrio furnissii* and some strains of *Vibrio damsela*), oxidase positive (except *Vibrio metschnikovii*), and have the ability to reduce nitrate to nitrite. Most are susceptible to the vibriostatic effects of the compound O/129.



FIGURE 2 There are over 200 serotypes of *Vibrio cholerae* which share a common flagellar H antigen and are distinguished from each other by a somatic O antigen. Serotype O1 and O139 are the strains associated with cholera. Image courtesy of J.B. Goforth, G. Hou, and E. Karatan. ©2011, Appalachian State University. All rights reserved. [doi:10.1128/microbiolspec.IOL5-0008-2015.f2](https://doi.org/10.1128/microbiolspec.IOL5-0008-2015.f2)

The pathogenic vibrios can be divided into two groups based on their ability to grow in a saline environment. Nonhalophilic vibrios grow in the presence or absence of sodium chloride and include *V. cholerae*, *V. cholerae* non-O1, and *Vibrio mimicus*. Halophilic vibrios, on the other hand, require sodium chloride to support growth and survival and reach very high concentrations in waters of 5 to 8% salinity.

Isolated vibrios can occasionally be confused with other bacteria of medical importance such as *Enterobacteriaceae*, *Pseudomonas*, *Aeromonas*, and *Plesiomonas*. The *Enterobacteriaceae* are straight rather than curved and are oxidase-negative with peritrichous or circumferential flagellae. *Pseudomonas* species, although oxidase-positive, have an oxidative rather than fermentative metabolism. Species of *Aeromonas* and *Plesiomonas* do not require sodium chloride for growth and are able to grow in the presence of vibriostatic O/129.

Most standard laboratory media used for biochemical testing contain 0.5% sodium chloride and therefore support the growth of both halophilic and nonhalophilic vibrios. Isolation is usually accomplished through selective or enrichment media. Alkaline peptone broth is the most suitable general enrichment media for all pathogenic *Vibrio* species. A modified two-step method has been successfully utilized to prevent bacterial overgrowth by other bacteria in the peptone broth. Thio-sulfate-citrate-bile salts-sucrose agar is the most widely used selective agar medium for isolation of pathogenic *Vibrio* species, but several newly described pathogens

may fail to grow on this agar medium. Furthermore, individual variations of commercially available thiosulfate-citrate-bile salts-sucrose agar may affect recovery of the organism.

Vibrios are aquatic organisms that can be found in a wide variety of environmental water sources such as oceans, estuaries, lakes, and ponds. The highest concentrations are generally achieved in the marine waters along the East and Gulf Coasts, primarily in the summer months; lower concentrations exist along the West Coast. Their numbers fluctuate widely, with marked variation due to such variables as temperature, salinity, sediments, and the presence of certain marine organisms, particularly the copepods and other plankton in which the vibrios may play a role in salt retention.

Water temperature appears to be the single most important variable affecting the growth and survival of the vibrios. Pathogenic vibrios are usually isolated from waters where temperatures exceed 10°C for at least several consecutive weeks. They are less frequently found in waters where temperatures exceed 30°C. Variation occurs worldwide and among species, *V. cholerae*, for example, prefers temperatures between 20 and 35°C.

Individual *Vibrio* species have different optimal sodium chloride requirements, with a range of 5 to 30%, which accounts for the primary isolation of these organisms from marine and estuarine waters. Although *V. cholerae*, *V. cholerae* non-O1/non-O139, and *V. mimicus* do not require sodium chloride for growth, they achieve higher numbers in its presence. *V. cholerae* has an optimal requirement of 2 to 20% salinity, whereas halophilic vibrios usually achieve optimal concentrations in sodium chloride concentrations of 5 to 8%. Pathogenic vibrios may be isolated from freshwater where salinity is less than 5% probably due to a complex interaction between high water temperatures and increased organic content which may compensate for the detrimental effects of low to absent salinity. Evaporation of fresh and brackish waters during summer months may increase the sodium chloride content.

As temperatures drop below 10°C, the pathogenic vibrios rapidly disappear from the water but can persist throughout the winter in the sediment. This has been shown for *V. parahaemolyticus*, *V. cholerae*, and *V. alginolyticus* and may well hold true for all pathogenic vibrios. As water temperatures increase during the spring and summer months, the organisms can then re-emerge to reach high water concentrations, accounting for as much as 26 to 40% of the total bacterial population in some areas.

By associating themselves with higher organisms such as shellfish, plankton, and fish, the vibrios may maintain high numbers and prolong their existence. Adsorption onto the chitinous component of plankton has been shown to significantly prolong the survival of some pathogenic vibrios, and it is possible that this represents a major means of prolonging survival for all pathogenic species. Bivalve molluscan shellfish which filter-feed on zooplankton may themselves become rapidly contaminated during periods of high bacterial counts. Improper storage of the shellfish may then allow proliferation of the pathogenic bacteria, with resultant outbreaks of food poisoning. Crustacean shellfish and fish can also become contaminated during such periods of abundant vibrio growth. *Vibrio* species have also been cultured from the teeth, skin, and gum lines of sharks.

The spectrum of human disease due to the pathogenic vibrios is mainly dependent on the causative species and ranges from mild self-limiting gastroenteritis and soft tissue infections to severe necrotizing wound infections and fulminant bacteremia, primarily in patients with underlying diseases (Table 2). Illness can result from a variety of means such as ingestion of contaminated shellfish or exposure of open wounds to contaminated seawater. Raw oyster eaters, particularly those with liver disease, have been shown to be at risk of developing vibrio illness. An increase in the incidence of vibrio-related illness is occurring for a variety of reasons including increased awareness by the public and medical profession of vibrio infections, improved biochemical and serologic means for the detection of vibrios, increased recreational exposure to coastal regions, increased foreign travel, increased seafood ingestion, and enhanced survival in immunocompromised individuals.

Vibrio species have been isolated from virtually every geographic region within the United States, although most cases occur along coastal areas. Over a 10-year period in a Gulf Coast community, 23 cases of *Vibrio* infections were reported, with *Vibrio vulnificus*, *V. parahaemolyticus*, and *V. cholerae* non-O1 accounting for all but 2 cases; disease included gastroenteritis in 3 patients, wound infection in 14 patients, and bacteremia in 12 patients. In 2012, 944 isolates were reported to the CDC from various human anatomical sites, 85% of them belonging to one of four species: *V. parahaemolyticus*, *V. alginolyticus*, *V. vulnificus*, and *V. cholerae* (excluding toxigenic O1 and O139), with gastrointestinal symptoms predominating. The CDC estimates that 8,028 *Vibrio* infections and 57 *Vibrio*-related deaths occur annually in the United States.

TABLE 2 Clinical presentations of pathogenic *Vibrio* infections in humans^a

Pathogen	Symptom			
	Diarrhea (watery/dysentery)	Wound	Otitis	Sepsis
Nonhalophilic				
<i>Vibrio cholerae</i> O1	+	±		
<i>V. cholerae</i> non-O1/non-O139	+/+	+	+	±
<i>Vibrio mimicus</i> ^b	+/+		+	
Halophilic				
<i>Vibrio parahaemolyticus</i>	+/+	+	±	±
<i>G. Grimontia hollisae</i> (EF13) ^b	+/±			±
<i>Vibrio fluvialis</i> (EF6)	+/+			
<i>Vibrio furnissii</i>	+			
<i>Vibrio alginolyticus</i>		+	+	
<i>Vibrio vulnificus</i> (L+) ^b	±			++
<i>Photobacterium damsela</i> (EF5)		+		
<i>Vibrio metschnikovii</i> (gp16)		+		±

^aSingle cases of *Vibrio cincinnatiensis* and *Vibrio harveyi* infection have been reported, with sepsis and shark bite wound infections, respectively.
^bEspecially associated with oyster consumption.

V. cholerae

Seven cholera pandemics have been recorded since 1817, six of which began before the 20th century. The seventh one began in Sulawesi, Indonesia, in 1961 and then spread to Asia, Africa, the Middle East, Oceania, and parts of Europe, while sparing the western hemisphere. No cases of domestically acquired cholera were reported in the United States after 1911 until 1973 when it was diagnosed in a resident of the Gulf Coast of Texas. In 1978, 11 people were involved in an outbreak of cholera following the ingestion of crabs gathered from a Louisiana coastal marsh, and in 1981, 16 oil workers were involved in a cholera outbreak on a Texas oil rig after eating rice cooked in water which had been contaminated by canal water containing sewage discharged from the rig. Between 2001 and 2011, 111 cases of cholera were reported to the CDC. Of these patients, 90 (81%) acquired cholera outside the United States, while 20 (18%) acquired it domestically (one case was lost to follow-up). Ten of the domestic cases consumed seafood from the Gulf Coast, six consumed non-Gulf Coast domestic seafood, three consumed imported seafood, and one had no history of seafood consumption. Toxigenic *V. cholerae* was confirmed by isolation in 108 of the 111 total isolates, and all but 1 were serogroup O1 (only 1 isolate of *V. cholerae* serogroup O139 was isolated, and that was a domestically acquired case involving seafood imported from an Asian supermarket). All isolates from this 10-year time span were biotype El Tor, 21% were serotype Inaba, and 79% were serotype Ogawa. All of the Gulf Coast isolates were of the El Tor biotype, serotype Inaba, which strongly suggests

that this organism is endemic along the Gulf Coast of Mexico.

V. cholerae serogroup O1 has been isolated from U.S. coastal waters of the Gulf of Mexico and Chesapeake Bay throughout the year, including toxin-producing strains. Numbers tend to be highest in the warmer summer months, and they are frequently associated with plankton and shellfish. Toxigenic and nontoxigenic strains of *V. cholerae* O1 have been cultured from shellfish harvested from U.S. waters. Pollution does not appear to be a necessary factor because the organism can be found in waters with no evidence of human waste. Case-controlled studies of localized outbreaks of *V. cholerae* O1 in the United States have shown that the recent ingestion of raw or partially cooked seafood or contact with contaminated water is a significant risk factor. Contaminated imported food can be a source of localized outbreaks in the United States. In 1994, a cluster of cases that occurred in Indiana was traced to food imported from El Salvador.

In January 1991, toxin-producing strains of *V. cholerae* O1, biotype El Tor, serotype Inaba appeared in several cities in Peru, which marked the first time in the 20th century that cholera had been reported in South America. In Haiti, an outbreak of cholera claiming over 8,000 lives has been ongoing since the earthquake of 2010. This outbreak has spread to South America, Mexico, and Cuba and is currently heading north toward the United States. Globally, an estimated 2.8 million cases and 91,000 deaths due to cholera occur annually in endemic areas, while another 87,000 cases with 2,500 deaths occur in nonendemic areas.

Over 200 serotypes of *V. cholerae* exist based on a somatic “O” antigen. These strains are phenotypically indistinguishable and share a common flagellar “H” antigen. *V. cholerae* serotype O1 is the strain most commonly associated with cholera, whereas a second serotype, O139, emerged in 1992 as an etiologic agent of cholera in Asia. All others are referred to as non-cholera vibrios, *V. cholerae* non-O1/non-O139, *V. cholerae* non-O1, non-O139, or nonagglutinable vibrios due to their inability to agglutinate in O1 or O139 antiserum. Most strains of *V. cholerae* O1 and O139 produce an enterotoxin (cholera toxin) which is responsible for the severe fluid losses seen in cholera. Non-O1 and non-O139 *V. cholerae* strains do not produce cholera toxin but may produce other toxins capable of eliciting an illness identical to cholera.

V. cholerae can be divided into two biovars, Classical and El Tor. Classical strains tend to be more virulent than the El Tor biotypes. El Tor strains were originally detected by their ability to lyse sheep erythrocytes, but this trait has not been shown to be consistent. They agglutinate chicken erythrocytes, are not sensitive to polymyxin B, and are Voges-Proskauer–positive. Classical strains fail to lyse sheep erythrocytes or agglutinate chicken erythrocytes, are Voges-Proskauer–negative, and are sensitive to polymyxin B. Biotypes may also be distinguished by phage susceptibility, a tool mainly used for epidemiologic purposes. Additionally, both biotypes can be further classified into one of three serotypes; Inaba, Ogawa, or Hikojima. The El Tor biotype initially appeared in cholera-endemic regions in the Ganges Delta in 1969 and quickly became the dominant biotype for most of the world. The more severe Classical strain has since reappeared in Bangladesh in 1982 and has become the dominant strain in that region. To date, the overwhelming majority of all biotypes in the western hemisphere have been of the El Tor biotype. Reasons for the persistence of El Tor strains include its greater ability to survive in the environment and a larger ratio of symptomatic cases to asymptomatic carriers (1:30–100 for El Tor and 1:2–4 for Classical biotypes).

Under adverse environmental conditions, *V. cholerae* O1 can enter a state of dormancy in which the organism remains viable and potentially pathogenic yet fails to grow on conventional culture medium. The presence of the organism under such conditions has clearly been demonstrated by fluorescent and immunologic techniques in the absence of a positive culture. It has been proposed that in this way, *V. cholerae* O1 may persist indefinitely and go undetected in waters such as the Gulf Coast, only to emerge periodically under more favorable

conditions to cause disease. This may also explain the periods between epidemics in cholera-endemic areas. This phenomenon has been referred to as “viable but non-culturable.”

In the United States, cases of cholera have usually occurred in the summer or fall months following the ingestion of raw or undercooked shellfish. Once in the intestinal tract, the organism may adhere to the intestinal cell wall by means of a specialized pilus, where it may grow and produce enterotoxin. Factors diminishing transit time, such as the ingestion of solid foods rather than liquids, may increase the chances of colonization and development of cholera. Furthermore, because the organism is less likely to survive in an acidic environment, the use of antacids or previous gastrectomy may increase the chances of developing disease. Adherence of the organism to chitin particles in shellfish may also enhance survival in acidic environments. For some reason, people with blood type O have been noted to be at a higher risk for cholera.

Cholera toxin is a heat-labile protein produced by most strains of *V. cholerae* O1 and O139. The molecule is composed of five B subunits arranged in a circle around an A1 and an A2 subunit. The B subunits are responsible for binding of the toxin to the receptor, ganglioside GM1, on cell membranes. The A1 subunit, bound to the complex by the A2 subunit, stimulates adenylate cyclase activity causing increased levels of cyclic AMP and hypersecretion of chloride, with the result of massive losses of salt and water. Cholera victims may lose over 1 liter of fluid per hour and up to 10% of their body weight in 4 to 7 days of diarrhea as a result of the toxin. Cholera toxin is not the only factor capable of causing the severe diarrhea in cholera victims; an identical illness has been noted in individuals infected with nontoxigenic variants of *V. cholerae* O1 and O139. Factors other than cholera toxin which have been proposed to play a role in producing illness include heat-stable toxins similar to toxins produced by *V. parahaemolyticus* and enterotoxigenic *E. coli*, lecithinase, phospholipase, and prostaglandin. *E. coli* may produce a plasmid-mediated toxin that is very similar in structure and mode of action to cholera toxin.

The incubation period in cholera varies between 6 hours and 5 days. The average is 2 days in endemic areas. Initial symptoms may include anorexia, abdominal cramping, and mild diarrhea. Vomiting without nausea typically begins within hours of the onset of diarrhea. Fever is usually absent or low-grade. Stools are initially brown in color and loose, but within hours become watery and pale gray in appearance and lose

all odor except for perhaps a “fishy” smell. Scattered flecks of mucous give the stools a “rice water” appearance. A feeling of relief, rather than tenesmus, accompanies each bowel movement. Peak stool losses occur around 24 hours of illness. Shock may occur within 12 hours in untreated cases, with death ensuing in 18 hours to 5 days. Gallbladder disease has been reported, including one case of acute cholecystitis in Alabama in which *V. cholerae* O1 was isolated from the gallbladder and bile. Serum vibriocidal antibodies were present as well. Chronic, asymptomatic gallbladder carriage has also been documented, especially in endemic areas of the world where the risk of exposure is great. Extraintestinal infections are rare; occasionally, nontoxigenic strains may be isolated from wound infections.

Cholera is fatal in less than 1% of properly recognized and treated cases, although it can be as high as 50% if untreated. During a typical outbreak in an endemic area, 75% of cases are asymptomatic, 20% are mild to moderate, and 2 to 5% are severe forms such as cholera gravis. The mainstay of therapy consists of rapid and effective fluid and electrolyte replacement. Oral rehydration is effective for most patients who are able to tolerate this mode of therapy. Currently, oral rehydration salt packets are available as well as a variety of other premixed rehydration solutions which have been shown to be effective in the treatment of cholera. In the event that vomiting is severe enough or the patient is too obtunded to tolerate enteral hydration, intravenous infusion of lactated Ringer’s solution has been effective in initial rehydration, with oral therapy being initiated as soon as the patient is able to tolerate it. Normal saline is less effective because it lacks bicarbonate and potassium. The amount of fluids administered should be determined on the basis of dehydration at presentation as well as the rate of ongoing losses.

Antibiotics may decrease the duration of illness, the period of vibrio excretion, and the amount of fluids needed for rehydration. It is usually indicated for severely dehydrated patients who are over 2 years of age. When indicated, antibiotics are administered orally (preferably) for 3 to 5 days. Effective first-line agents include ciprofloxacin, doxycycline, tetracycline, and furazolidone. Doxycycline, as a single dose of 1,000 mg, is as effective as multiple doses of tetracycline, so it has become the preferred antibiotic of choice when antibiotics are used. A single dose of ciprofloxacin 1,000 mg, is also effective. Azithromycin is recommended for children and pregnant women. It should be noted that fluid rehydration is still considered to be the mainstay of therapy. Antispasmodics, antidiarrheal agents, and corticosteroids are not

indicated in the treatment of cholera. Two oral vaccines are available, but they are not available in the United States and offer incomplete protection. They should not be used to replace standard prevention and control measures.

Non-O1/Non-O139 *V. cholerae*

Strains of *V. cholerae* that do not agglutinate in O1 or O139 antisera, or non-O1/non-O139 *V. cholerae*, are common inhabitants of both sewage-contaminated and sewage-free waters of bays, estuaries, brackish inland lakes, and seafood. Previously referred to as non-O1 *V. cholera*, they became known as non-O1/non-O139 after *V. cholerae* O139 was identified in Asia in 1992 and subsequently found to be a causative agent of cholera. Besides being environmental contaminants, the organism has been isolated from domestic animals, waterfowl, and a variety of wildlife. Non-O1/non-O139 species have even been implicated in enteric infections of horses, lambs, and bison in western Colorado. Most disease-associated isolates have been obtained from the coastal waters of Florida, the Gulf of Mexico, and the Chesapeake Bay, although isolates may be found all along the East Coast. Fewer isolates are noted along the West Coast, possibly due to colder water temperatures. Infections due to non-O1/non-O139 strains have also been acquired from freshwater lakes distant from the sea. Infections are more common during the warmer-water months of summer and fall.

The first reported case of human disease due to non-O1/non-O139 strains of *V. cholerae* acquired in the United States occurred in 1972 in Louisiana in an individual who developed profuse and prolonged diarrhea following the ingestion of raw oysters. Since then, this organism has been increasingly recognized as a cause of human illness in this country, usually occurring as cases of sporadic illness. In a Chesapeake Bay hospital over a 15-year period, 40 *Vibrio* isolates were obtained, of which 10 were strains of non-O1/non-O139 *V. cholera*. Similarly, in a Gulf Coast community over a 10-year period, 4 of 23 *Vibrio* isolates were non-O1/non-O139 *V. cholerae*. Although diarrheal illness is the most common manifestation of disease due to non-O1/non-O139 *V. cholerae* strains, 10% of the cases are wound infections, and an additional 10% are ear infections. Septicemia, meningitis, and acalculous cholecystitis have been reported. Unlike the strain which causes cholera, non-O1/non-O139 species are rarely linked to epidemic disease. Isolated outbreaks of illness have occurred usually in association with a common contaminated food source. Three previous outbreaks of diarrheal illness

linked to non-O1/non-O139 *V. cholerae* in Czechoslovakia, the Sudan, and Australia were linked to food sources in two (potatoes and an asparagus salad) and polluted well water in the other. Around 40 cases of non-O1/non-O139 *V. cholerae* have been reported to the CDC every year since 2000. Usually, cases are seasonal and tend to peak in the late summer and early fall when water temperatures are the warmest.

More than half the non-O1/non-O139 *V. cholerae* isolates received by the CDC are from stool samples. Gastroenteritis is the most common manifestation of illness in the United States and is almost always due to the ingestion of raw oysters. In the United States, the incubation period has generally been less than 48 hours, but it was as long as 4 days in at least one foreign outbreak. The following symptoms were noted in one review of U. S. cases: diarrhea (100%), abdominal cramps (93%), fever (71%), and nausea and vomiting (21%). Another review reported nausea and vomiting occurring more frequently (77% and 69%, respectively). The diarrhea can be severe, with up to 30 watery stools per day, and up to 25% of cases have had bloody diarrhea. In some cases fluid losses may equal that seen in cholera. Illness generally lasts an average of 6.4 days in the cases reported in the United States (range 2 to 12 days) but has been less than 2 days in some overseas outbreaks.

Non-O1/non-O139 strains of *V. cholerae* exert their pathogenic effects through a variety of extracellular toxins and hemolysins. Enterotoxins identical or nearly identical to cholera toxin have been detected. Although rare in the United States, up to 40% of isolates in India and Bangladesh may produce this toxin. In fact, severity of diarrheal illness seems to correlate with production of this toxin. Cholera toxin-like producing strains of non-O1/non-O139 *V. cholerae* are capable of causing outbreaks of cholera-like illness (*V. cholerae* O141 and O75 in the United States and *V. cholerae* O6, O10, O14, and O37 in the rest of the world). Toxigenic strains of non-O1/non-O139 *V. cholerae* are derived from non-pathogenic strains and are facilitated by the exchange of multiple genes involved in pathogenicity. Other factors which may be produced include enterotoxins other than the cholera-like toxin including an enterotoxin that is also produced by *V. mimicus* and *Vibrio fluvialis*, several hemolysins including a hemolysin similar to the Kanagawa hemolysin produced by *V. parahaemolyticus*, a Shiga-like toxin which may be responsible for the occasional bloody diarrhea, repeat-in-toxins (now called MARTX toxins in *V. cholerae*) which facilitate small intestinal colonization, and type III secretion systems

such as those found in *V. parahaemolyticus* and other Gram-negative enteric pathogens.

Septicemia due to non-O1/non-O139 strains of *V. cholerae* is a less frequent complication of infection. During 1977 through 1979, 70 isolates of *V. cholerae* were reported to the CDC, of which 12 were obtained from the blood. Heavy encapsulation by these isolates has been associated with their virulence in septicemia. The case fatality for septicemia has been estimated at 61.5% based on previously reported cases in the United States. Predisposing factors include alcohol abuse, previous gastric surgery, advanced age, and chronic underlying conditions such as hematologic malignancy, liver disease, immune deficiency, diabetes mellitus, peripheral vascular disease, and achlorhydria. The source of the vibrio in many cases is raw oyster ingestion. In addition to septicemia and gastroenteritis, non-O1/non-O139 *V. cholerae* has also been implicated in cases of wound infections, otitis media and otitis externa, prostatic abscess, cholecystitis, pneumonia, and meningitis, including one case of neonatal meningitis in an infant who drank milk from a bottle stored near live crabs. Extraintestinal infections are usually the result of contact with seawater.

Most cases of non-O1/non-O139 *V. cholerae* are relatively mild cases of gastroenteritis which are usually self-limiting in nature; only a minority require hospitalization. Treatment should consist of supportive care with intravenous hydration in severe cases of gastroenteritis. Septicemic cases and severe localized infections such as meningitis should be treated with intravenous antibiotics. In cases of severe gastroenteritis, antibiotics may decrease the severity of illness and shorten its duration. Non-O1/non-O139 *V. cholerae* strains are susceptible *in vitro* to a wide range of antibiotics including tetracycline, trimethoprim-sulfamethoxazole, chloramphenicol, and nalidixic acid. Susceptibilities to ampicillin and gentamicin are variable.

V. parahaemolyticus

V. parahaemolyticus has a worldwide distribution in both tropical and temperate inshore coastal and estuarine waters. Isolates have been obtained from water, sediment, suspended particulates, plankton, fish, and shellfish in a marine environment. Although halophilic, the organism can occasionally be isolated from freshwater areas, possibly through association with the chitin of plankton and shellfish or sediments which may allow it to survive in areas of lower salinity. Most cases of illness occur in summer when warmer water temperatures favor growth of the organism. During winter,

the organism can be isolated from sediment. *V. parahaemolyticus* has been isolated from nearly every coastal state in the United States and is frequently found in Canadian waters as well. Outbreaks of diarrheal illness are the most common form of disease caused by this organism, and this usually follows the ingestion of raw or improperly cooked seafood, particularly crabs, shrimp, lobsters, and raw oysters. Most U.S. outbreaks have been caused by gross mishandling of seafood, such as improper refrigeration, insufficient cooking, cross-contamination, and recontamination. Incubation of the organism in seafood can reduce the generation time to as low as 12 minutes. Previously thought to be a rare strain of *V. parahaemolyticus* and not described in the United States until 1982, a urease-positive strain of *V. parahaemolyticus* representing a new serovar has been established as the predominant cause of *V. parahaemolyticus*-associated gastroenteritis on the west coast of the United States and Mexico.

In Japan over 70% of food-borne diarrheal illness is caused by *V. parahaemolyticus*. First described as a pathogen in the United States in 1971 following the ingestion of improperly cooked crabs in Maryland, the organism has been increasingly recognized as a cause of sporadic cases of diarrheal illness, usually in association with food-borne outbreaks. During 1973 to 1998, 40 outbreaks of *V. parahaemolyticus* were reported to the CDC, with most cases occurring in the warmer months. In a Chesapeake Bay hospital over a 15-year period, *V. parahaemolyticus* was the most common *Vibrio* species identified and accounted for 16 of 40 isolates obtained from 32 patients. During an outbreak of *Vibrio* gastroenteritis among attendees at a scientific congress in New Orleans, *V. parahaemolyticus* accounted for 35 of the 51 stool specimens yielding a *Vibrio* species. In 1997, the largest reported outbreak of *V. parahaemolyticus*-confirmed infections occurred in North America. A total of 209 cases were identified, with one fatality. The outbreak was associated with raw oysters harvested from California, Oregon, Washington, and British Columbia. Extraintestinal manifestations may also occur, including sometimes fatal septicemia. Infections are usually seasonal, with 70% of infections occurring between May and October when water temperatures are warmer. Large outbreaks due to the ingestion of raw oysters occurred in the United States in 1997, 1998, and 2006.

The incubation period has ranged from 4 to 96 hours. In a review of eight *V. parahaemolyticus* outbreaks in the United States, the clinical manifestations included diarrhea (98%), abdominal cramps (82%), nausea (71%), vomiting (52%), headache (42%), fever (27%),

and chills (24%). The illness was self-limited in most cases, with a median duration of three days. Fever rarely exceeds 38.9°C. Abdominal pain can be severe. The diarrhea is acute in onset and usually watery and mild, although it can rarely be severe enough to cause dehydration, hypotension, and acidosis. A dysentery-like illness with fecal leukocytes, superficial ulcerations of the colonic mucosa, and blood and mucous in the stool has been described in India and Bangladesh. This form of disease has rarely been encountered in the United States. The incubation period in dysentery-like disease is shorter (as short as 2.5 hours), although the duration of illness approximates that of the more common form of illness.

The ability to cause human disease has been associated with a heat-stable enterotoxin capable of lysing erythrocytes on Wagatsuma agar. Strains producing this thermostable direct hemolysin (Vp-TDH) are known as “Kanagawa positive,” named after the prefecture in Japan where it was first studied. The observation that over 95% of clinical isolates and less than 1% of environmental isolates are Kanagawa positive has suggested that Vp-TDH is the toxin associated with pathogenicity. In several studies involving animal models, purified Vp-TDH was able to produce clinical and histopathologic effects similar to those seen in disease due to *V. parahaemolyticus*. Furthermore, absence of the gene for Vp-TDH results in loss of enterotoxicity. In one report from the Pacific Northwest, only 6 of 13 clinical isolates from patients with diarrhea or wound infections due to *V. parahaemolyticus* were Kanagawa positive. The ability of Kanagawa negative strains lacking Vp-TDH to produce diarrheal illness has led to the discovery of other enterotoxins. A second group of hemolysins, known as Vp-TDH-related hemolysins, or Vp-TRH, are Kanagawa negative but urease positive and tend to cluster along the Pacific coast in the United States as well as parts of Asia. Almost all of these strains carry the gene for Vp-TRH. In addition to Vp-TDH and Vp-TRH, other unclarified virulence factors may play a role in pathogenicity. Up to 6% of clinical isolates lack genes for either Vp-TDH or Vp-TRH. A Shiga-like toxin has been identified as well as a type III secretion mechanism, such as that found in other enteric pathogens, which may explain more invasive diarrhea.

In rare instances *V. parahaemolyticus* can result in extraintestinal infections. A history of trauma or insult to the infected anatomical site can be elicited in the majority of cases. Wound infections or cellulitis can develop as a primary focus of infection or can result from secondary hematogenous seeding. Although less common than stool isolates, wound isolates can constitute

a significant number of *V. parahaemolyticus* isolates. Out of 16 *V. parahaemolyticus* isolates collected over a 15-year period in the Chesapeake Bay region, 5 were acquired from wounds. Vascular thrombosis and gangrene have been reported. In some instances, *V. parahaemolyticus* and another *Vibrio* species may be isolated concurrently. Ocular and ear infections can occur, as well as pneumonia and osteomyelitis.

Most cases of *V. parahaemolyticus* gastroenteritis are self-limited and resolve in a matter of days. Rarely are cases severe enough to require vigorous fluid support. Mortality is unusual and has been estimated to be 0.04% in Japan. From 1981 to 1988, there were four fatalities due to *V. parahaemolyticus* in Florida. All patients were bacteremic and had either cirrhosis or an underlying malignancy. In view of the fact that most cases of infection due to *V. parahaemolyticus* are gastroenteritis, methods to avoid such illness include the proper handling of seafood. Heating at 60°C for 15 minutes kills *V. parahaemolyticus*. In addition, storing at temperatures at or below 4°C inhibits growth of the organism. People susceptible to septicemia such as those with liver disease or any other immunosuppressive condition should probably avoid raw seafood ingestion.

Vibrio vulnificus

Beginning in 1964, the CDC began receiving extra-intestinal isolates that were thought to be variants of *V. parahaemolyticus* but were shown to be different by means of a variety of biochemical tests including the ability to ferment lactose. They were referred to initially as the “halophilic lactose-positive marine vibrio”; the name *Beneckea vulnifica* was initially proposed although not widely accepted. The virulence of this species was first recognized in 1976 by a review of clinical isolates reported to the CDC, which revealed that 53% of the isolates were recovered from blood. The name *Vibrio vulnificus* was formally recognized in 1979. Three biotypes exist: biotype 1, which is associated with human infections, biotype 2, which is an eel pathogen, and biotype 3, which is a hybrid of the other two biotypes and has been associated with human wound infections associated with tilapia in Israel. This organism has been found in seawater, sediments, zooplankton, and shellfish. In two studies, over 50% of oysters sampled during selected months and 11% of crabs harvested during summer months yielded *V. vulnificus*. Water temperature seems to be an important factor; organisms are rarely isolated from waters with temperatures lower than 17°C. Almost all cases of infection have been reported between the months of May through October.

Cases have been reported along both the Pacific and Atlantic coasts (as far north as Cape Cod), Hawaii, the Gulf of Mexico, and occasionally in such inland areas as New Mexico, Oklahoma, Kentucky, and the Great Salt Lake.

In a Gulf Coast community over a 10-year period, 12 of 23 *Vibrio* isolates obtained were *V. vulnificus*, of which 9 were wound isolates. Similarly, in a Chesapeake Bay Hospital over a 15-year period, 10 of 40 *Vibrio* isolates were *V. vulnificus*, of which 7 were obtained from wounds. *V. vulnificus* accounts for approximately 8% of the aerobic bacteria in the Chesapeake Bay. In the two months following Hurricane Katrina in 2005, 14 cases of *V. vulnificus* wound infections were reported in the Gulf region, of which 3 were fatal. The estimated incidence of *V. vulnificus* infections in the United States is approximately 0.5/100,000 population per year, but data from surveillance shows this to be much lower at 0.4 to 0.5/100,000 population per year. However, rates have been rising dramatically since 1996. The organism can proliferate in seafood at room temperature but is killed by storing at or near freezing temperatures or by cooking seafood at boiling temperatures. *V. vulnificus* has been isolated in oysters that have been refrigerated for 4 days.

V. vulnificus is one of the most invasive and rapidly lethal human pathogens ever described. Two major syndromes can result from infection with this organism. Primary septicemia typically follows the ingestion of raw oysters by individuals with liver disease. This syndrome can have a rapidly fatal course in up to 60% of cases. The other major presentation is that of wound infections, which may occur by either primary inoculation or secondary hematogenous spread in a bacteremic individual. Antibiotics, vigorous debridement, and occasionally amputation are necessary to control the massive necrosis and systemic spread which can occur (Fig. 3). Unlike other vibrioses, gastroenteritis is not a major hallmark of infection with this species, and although gastrointestinal symptoms may accompany other forms of illness, the relationship is unclear. In one epidemiologic study of *V. vulnificus* infections in Florida from 1981 to 1987, 7 out of 62 patients (11%) had gastrointestinal symptoms as their only manifestation of disease. Stool specimens yielded *V. vulnificus*, and blood cultures were negative. The diarrhea was described as watery, profuse, and accompanied by vomiting and abdominal pain. Six patients (86%) were hospitalized for a median of 6 days. Medications that reduce gastric acidity may be a factor in the development of gastroenteritis. Of 62 cases reported in Florida between 1981



FIGURE 3 Wound infections due to *Vibrio vulnificus* may occur as a result of primary inoculation or hematogenous spread in a bacteremic individual. Prompt debridement, appropriate antimicrobial agents, and occasionally amputation, are necessary to avoid severe tissue necrosis and fulminant sepsis. Image from R.A. Schwartz 2015 *Vibrio vulnificus* Infection (Medscape). doi:10.1128/microbiolspec.IOL5-0008-2015.f3

and 1987, 38 were primary septicemia (62%), 17 were wound infections (27%), and the remaining 7 were the gastrointestinal illness described above (11%). Other infections due to *V. vulnificus* include pneumonia, endocarditis, osteomyelitis, ocular infections, and meningitis. One reported case of endometritis due to *V. vulnificus* occurred in a female who had engaged in sexual intercourse in seawater.

The severity of infections due to *V. vulnificus* depends on host as well as bacterial factors. The presence of an acidic polysaccharide capsule correlates strongly with virulence. This capsule may confer resistance to phagocytosis and bactericidal activity of human serum. Strains may shift between encapsulated and unencapsulated forms at a very low frequency by poorly understood mechanisms. Unencapsulated forms taken up by oysters show a high rate of transition to the encapsulated form, which suggests that passage in the oyster selects toward the virulent encapsulated form of the organism. Growth of both encapsulated and unencapsulated phenotypes is enhanced significantly by the presence of iron. The organism is able to use transferrin-bound iron for growth if the transferrin is 100% iron-saturated (normal human serum is 30% saturated). Iron in hemoglobin and hemoglobin-haptoglobin complexes may also be used. *V. vulnificus* also produces siderophores, which are low-molecular-weight chelators that bind available iron. Mouse studies have shown that passage of the organism through the animal may enhance virulence

significantly, which suggests that the reintroduction of strains shed by infected individuals may increase the potential pathogenicity of environmental or food-borne strains if reintroduced to the environment by infected individuals. The organism may produce various toxins and enzymes. They include mucinase, protease, lipase, DNase, chondroitin sulfatase, hyaluronidase, cytotoxin, and collagenase. The contribution to virulence by each of these factors has not yet been fully determined. A repeat-in-toxin (RtxA1) has been identified that is similar to the one found in *V. cholerae*. RtxA1 appears to trigger excess generation of reactive oxygen species in the host cell, resulting in cell death. This appears to be a major mechanism of the virulence of *V. vulnificus*. Anticytolysin-directed antibody has been detected in the serum of individuals with invasive disease, suggesting that this toxin may also play a major role in invasive forms of disease.

Individuals susceptible to primary septicemia are most commonly those with liver disease, particularly alcoholic cirrhosis. Adults with liver disease who eat raw oysters are 80 times more likely to develop illness with *V. vulnificus* than those without liver disease, and 200 times more likely to die from *V. vulnificus* infections. This is due in part to shunting of blood around the liver, thereby bypassing the hepatic reticuloendothelial system and subsequent clearing by hepatic macrophages. Additionally, these individuals may also have deficiencies in leukocyte chemotaxis and complement that can impair host defenses. Patients with hepatic disease commonly have high serum iron levels due to liberation of iron stores from damaged hepatocytes. In addition, other conditions leading to increased serum levels of iron, such as thalassemia major, and hemochromatosis, can also contribute to primary septicemia. Alcoholics without liver disease are at risk for primary septicemia, probably due in part to saturated transferrin levels. Men, particularly older men, appear more susceptible than women. Other conditions predisposing to primary septicemia include hematopoietic disorders, chronic renal insufficiency, dyspeptic disease or a history of gastric resection, the use of immunosuppressive drugs, and diabetes mellitus. Primary septicemia occasionally occurs in previously healthy people.

There is little doubt that the gastrointestinal tract is the portal of entry in primary septicemia. The organism can survive between pH 3.6 and 12.5 when incubated at 37°C for 1 hour and grows best between pH 7 and 9. Therefore, the organism is capable of surviving passage through the stomach. Most commonly, this occurs following the ingestion of raw oysters. An epidemiologic

study in Florida of oyster eaters between 1981 and 1988 estimated the age-standardized annual incidence of any *Vibrio* illness per million is 95.4 for raw oyster eaters with liver disease, 9.2 for raw oyster eaters without liver disease, and 2.2 for non-raw oyster eaters. Although these estimates were based on the chances of developing any *Vibrio* illness, the risk was shown to be greatest for *V. vulnificus*. Other types of raw fish or shellfish may cause septicemia, and on previous occasions, illness followed ingestion of deep-fried fish, grilled crab, boiled shrimp, and broiled grouper. *V. vulnificus* invades the gastrointestinal mucosa at the level of the proximal small bowel and moves into the systemic circulation to produce sepsis. On occasion, the organism may penetrate the intestinal wall into the ascitic fluid of cirrhotics to produce peritonitis.

The mean incubation period is 16 hours, although it has been as long as 2 weeks. Primary symptoms associated with septicemia include fever (94%), chills (91%), and nausea (58%). Diarrhea occurs in less than half of patients with primary septicemia. Approximately one third become hypotensive within 12 hours of admission. Signs and symptoms of disseminated intravascular coagulation and septic shock may appear. Thrombocytopenia and leukopenia are common, although leukocytosis can appear less frequently. Arthritis and arthralgias may develop, and the organism has been cultured from affected joints. Other associated clinical characteristics may include the rapid development of anemia, adult respiratory distress syndrome, and heart block.

Over 70% of patients with primary septicemia develop skin lesions, usually within the first 36 hours of illness. Lesions are more common on the trunk and extremities and frequently begin as tender erythematous or ecchymotic areas which may progress to bullae or vesicles that develop into necrotic ulcers. They may occasionally appear as ecthyma gangrenosa-like lesions, erythema multiforme, cellulitis, and papular or maculopapular eruptions. Similar lesions have been reported in individuals who had bacteremia due to *Pseudomonas aeruginosa*, *Aeromonas hydrophila*, and *Yersinia enterocolitica*. Gangrene of a limb can develop as a result of major vessel occlusion. Histological examination reveals cellulitis with subcutaneous tissue necrosis and septal panniculitis characterized by paucity of an inflammatory infiltrate in the dermis. Necrotizing vasculitis and subepidermal bullae, as well as Gram-negative coccobacilli, can be seen. Gram stain and culture of vesicular or bullous fluid may yield the organisms, which have been described as resembling seagulls.

Localized wound infections with *V. vulnificus* may occur in otherwise healthy people after an open wound comes in contact with seawater or seafood contaminated with the organism. A typical scenario is the development of a wound infection following injuries sustained while peeling shrimp, cleaning crabs, or shucking oysters. Infection may also follow exposure of a pre-existing wound with seawater. Approximately one third to one half of cases of wound infections due to *V. vulnificus* occur in patients who have underlying illnesses such as alcohol abuse, congestive heart failure, stasis ulcers, arthritis, liver disease, diabetes mellitus, and malignancy. Initially, the wound may appear trivial, but in a matter of hours, it characteristically becomes edematous and erythematous with development of lymphadenopathy and lymphangitis. Intense pain may develop at the wound site. Patients are frequently ill with fever and chills. Anorexia, nausea, and vomiting may occur, although less frequently than in primary septicemia. Occasionally, more than one *Vibrio* strain may be isolated from the wound. Unlike primary septicemia, wound infections usually remain localized, but bacteremia with secondary development of cutaneous infections can occasionally occur. Approximately one third of individuals with localized wound infections have positive blood cultures. Individuals with underlying diseases (rarely healthy people) may develop progressive cellulitis, myositis, or fasciitis. Leukocytosis is usually noted; thrombocytopenia and disseminated intravascular coagulation generally do not occur. Severe necrotizing fasciitis can occasionally occur with gross purulence and easy separation of fascial planes. Findings of fascial necrosis separates this entity from cellulitis. The histopathology of wound infections is similar to that which occurs in primary septicemia, although it is generally not as severe. Mortality ranges from 7 to 22% of cases, being higher in those individuals with underlying diseases, who are more likely to develop progressive soft tissue involvement and bacteremia.

Treatment of *V. vulnificus* infections consists of rapid recognition with prompt institution of antibiotics and supportive measures along with management of adult respiratory distress syndrome, disseminated intravascular coagulation, and shock, if present. Surgical debridement of all necrotic tissue is recommended. Proximal amputation of infected limbs may be necessary in severe cases of wound infection. Tetracycline has been shown to be highly effective against *V. vulnificus* in a mouse model. Low efficacy was noted for ampicillin, cefotaxime, and cefazolin. Carbenicillin and gentamicin were not effective. Current antibiotic regimens consist

of ceftazidime (2 g intravenously three times a day) and doxycycline (100 mg orally or intravenously twice daily) or doxycycline in combination with ciprofloxacin or an aminoglycoside. The data supporting fluoroquinolones is limited; ciprofloxacin has been used effectively in the treatment of one case of infection due to *V. vulnificus*, but conclusive clinical studies are lacking.

V. mimicus

In 1981, a new *Vibrio* species was detected via DNA and biochemical analysis. Previously thought to be biochemical variants of *V. cholerae*, the new species differed in its inability to ferment sucrose and its negative Voges-Proskauer reaction. The name *Vibrio mimicus* was proposed because of its similarity to *V. cholerae*. Current evidence suggests that *V. mimicus* and the *V. cholerae* O1 strain that was implicated in the sixth cholera pandemic likely diverged from a common ancestor. The organism is nonhalophilic and has been isolated from both saltwater and freshwater. A high percentage of strains (49%) are able to grow in 6% sodium chloride. Brackish water with an average salinity of 4.0‰ was found to be suitable for *V. mimicus*. It has been isolated from fish and shellfish (usually oysters), as well as freshwater prawns. Unlike *V. cholerae*, it probably does not adhere to plankton samples in the environment. Over the period 1977 through 1981, the CDC received 21 clinical isolates of this organism, although the true incidence is unknown. Of these isolates, 19 were obtained from stool samples, and the other 2 were obtained from human ears. Clinical isolates have been obtained from the waters of the Gulf of Mexico, the mid-Atlantic coast, and the Chesapeake Bay. Among 40 *Vibrio* isolates collected over a 15-year period in a Chesapeake Bay hospital, 3 were identified as *V. mimicus*; all were isolated from stool samples. The organism has also been implicated along with *V. fluvialis* in a case of terminal ileitis.

Gastrointestinal illness is the predominant manifestation of disease, and this typically follows the ingestion of seafood, primarily raw oysters. In the largest reported review of illness due to *V. mimicus* in the United States, involving 21 cases, the median incubation period was shown to be 24 hours with a range of 3 to 72 hours. Symptoms included diarrhea (94%); nausea, vomiting, and abdominal cramps (67%); fever (44%, occasionally up to 38.3°C); and headache (39%). Of the 17 patients with diarrhea, 3 had bloody diarrhea (18%). The median leukocyte count was 13,400, with a median differential of 73% polymorphonuclear cells, 7% bands, 16% lymphocytes, and 4% monocytes. Electrolytes were all

within normal limits. Diarrheal illness lasted a median of 6 days. Two cases were ear infections acquired from contact with seawater. Outbreaks of seafood-associated gastroenteritis caused by *V. mimicus* has been described in Japan. Of clinical isolates reported in the United States between 2004 and 2009, 55% were from hospitalized patients; however, in a large outbreak in Thailand linked to contaminated fish soup, illnesses were milder, with 91% reporting diarrhea, 66% abdominal pain, and 27% vomiting. Most symptoms resolved in less than 36 hours, with less than a third seeking medical attention.

Approximately 10% of clinical strains and 16% of all strains produce a heat-labile toxin that appears to be identical to cholera toxin. An enterotoxin similar to that described for non-O1 *V. cholerae* and *V. fluvialis* has been described, as well as a type III secretion mechanism similar that seen in *V. parahaemolyticus*.

Antimicrobial testing has shown that *V. mimicus* is susceptible to tetracycline, which may be the drug of choice in severe infections. Trimethoprim/sulfamethoxazole, aminoglycosides, chloramphenicol, and ampicillin have been shown to be effective *in vitro*. In one study, 88% of environmental isolates were shown to be resistant to ampicillin, with intermediate sensitivity and no resistance to tetracycline in 17%.

V. fluvialis

V. fluvialis was first isolated in 1975 from the stool of a patient in Bahrain with diarrhea. In 1976 to 1977, it was responsible for an outbreak of diarrheal illness involving over 500 people in Bangladesh. About one half of the patients were children under 5 years of age. Originally referred to as group F by the Public Health Laboratory in Maidstone, England, and enteric group EF-6 by the CDC, it was later named *V. fluvialis* (from the Latin “river”) in 1980 on the basis of its original isolation from river and estuarine waters. This organism has marked biochemical similarities to *Aeromonas* species and can be distinguished by its ability to grow in 6 to 7% sodium chloride (*Aeromonas* species do not grow) and its inability to grow in the presence of the vibriostatic agent O/129 (*Aeromonas* species do grow). Based on these similarities, it is probable that many clinical isolates previously reported as *Aeromonas* species may have in actuality been *V. fluvialis*. One third of organisms labeled as *Aeromonas* in the past by the British Public Health Laboratories were found on reexamination to actually be *V. fluvialis*.

In the United States, the organism has been isolated from water and sediment in New York Bay, shellfish in Louisiana, and water and shellfish in the Pacific

Northwest and Hawaii. Almost all cases of infection due to *V. fluvialis* in the United States have been gastrointestinal illness. A history of seafood ingestion prior to the onset of illness is reported in most cases. In Florida between 1982 and 1988, 12 clinical isolates of *V. fluvialis* were recovered: 10 were obtained from stool samples, 1 was obtained from the drainage of a colostomy bag, and 1 was recovered from a wound. One isolate was recovered from a stool sample in a Chesapeake Bay hospital over a 15-year period. A fatality occurring in the United States due to *V. fluvialis*-related illness has been reported in the case of a Texas man who developed profuse diarrhea with electrolyte imbalance. Outbreaks of gastroenteritis linked to a common food source have been reported. Occasionally, wound infections can occur in association with injuries sustained near seawater. The organism was recovered from purulent bile in one patient in Japan with acute suppurative cholangitis.

The median incubation period for gastrointestinal illness has been reported as 39 hours, with a range of 16 to 60 hours and a median duration of illness of 6 days (range 1 to 60 days). In the Bangladesh outbreak, reported clinical features included diarrhea (100%), vomiting (97%), abdominal pain (75%), moderate to severe dehydration (67%), and fever (35%). Invasive disease probably occurs, because 75% of the patients in that outbreak had fecal leukocytes and blood in the stools. Secondary infection occurs rarely.

V. fluvialis is capable of stimulating fluid accumulation in rabbit ileal loops. An enterotoxin similar to that described for non-O1/non-O139 *V. cholerae* and *V. mimicus* has been described. Other factors associated with toxicity may also be produced with a variety of effects including cytolytic activity against mammalian erythrocytes, lethal activity in mice, and nonhemolytic cytotoxicity.

Severe cases of gastroenteritis should be treated with intravenous fluid, electrolyte replacement, and antibiotics. *V. fluvialis* is sensitive to tetracycline, ampicillin, chloramphenicol, gentamicin, and trimethoprim/sulfamethoxazole.

Grimontia (Vibrio) hollisae

Vibrio hollisae, now known as *G. hollisae*, was originally known as enteric group EF-13 until 1982 when it was shown to be a separate species by DNA hybridization studies. It was named after the researcher at the CDC who first identified it. In 2003, it was later reclassified as *G. hollisae* due to phylogenetic differences with the vibrios. It grows inconsistently on thiosulfate-citrate-bile salts-sucrose agar and thus may not be

isolated routinely. Isolation may rely on recovery of colonies on blood agar plates. There have been few ecological studies of *G. hollisae*, but it has been isolated from deep sea invertebrates and healthy coastal fish. The CDC received 15 clinical isolates from 1971 to 1981, of which 14 were stool isolates and 1 was a blood isolate. Cases have occurred in states along the Atlantic and Gulf Coasts, with three cases in Florida, four in Maryland, and one each in Virginia and Louisiana. In Florida in 1981 to 1988, 34 isolates of *G. hollisae* were identified from clinical cases, of which 28 were obtained from individuals with gastrointestinal illness, 4 were from individuals with septicemia, and 2 were isolated from wounds. The risk of infection is usually associated with raw seafood ingestion, particularly raw oysters, although cases of illness have followed ingestion of fried fish and fish preserved by drying and salting, which suggests that this organism may be resistant to some methods of cooking and preservation.

In nine selected cases of gastrointestinal illness reported to the CDC, all had diarrhea and abdominal pain, five had vomiting, and five had fever. Diarrhea was bloody in one case. The median white blood cell count was 11,200 cells per μl . The median duration of illness was 1 day (range 4 hours to 13 days). Reported cases of septicemia include one fatal case occurring in an individual with hepatic cirrhosis and another case occurring in a 65-year-old man who developed septicemia following consumption of a freshwater catfish and was successfully treated with tobramycin, cefamandole, and tetracycline.

Several toxins may be produced, including a hemolysin with potential virulence activity and a heat-sensitive enterotoxin which has been associated with some virulent strains. *G. hollisae* is known to have a type III secretion mechanism as well as a thermostable direct hemolysin similar to that in *V. parahaemolyticus*.

Photobacterium (Vibrio) damsela

P. damsela was originally known as enteric group EF-5. It was renamed *V. damsela* in 1981 after the damselfish, for which it is an important pathogen. After several name changes, it is now known as *Photobacterium damsela*. It is known to cause skin ulcerations and death in fish and has been isolated from seawater. Between 1971 and 1981, eight clinical isolates were reported to the CDC, of which seven were obtained from wounds (one was a urine isolate). Cases occurred on the Atlantic, Pacific, and Gulf coasts and were reported in Florida, Louisiana, Hawaii, and the Bahamas. Typically, infection is the result of injuries sustained to the foot or

leg while swimming or handling fish. Infection is usually seasonal and probably dependent on water temperature and possibly its interaction with certain fish species.

Lesions usually began as erythematous and indurated areas which may later exhibit a purulent discharge. Immunocompetent patients may require little more than local wound care, but severe necrotizing and occasional fatal infections have been described. One fatal case involved a diabetic, alcoholic patient who had sustained a small laceration to his hand while cleaning catfish. The initial superficial wound evolved into an edematous and necrotizing process with bulla formation. He subsequently died of medical complications including disseminated intravascular coagulation. Tissue damage may be toxin mediated. *P. damsela* may produce an extracellular hemolytic toxin which has been shown to be lethal in mice. Of five *P. damsela* isolates tested, all were sensitive to gentamicin and chloramphenicol, four were sensitive to tetracycline and cephalothin; none were sensitive to sulfonamides, ampicillin, or penicillin.

V. alginolyticus

Originally classified as biotype 2 of *V. parahaemolyticus*, this organism was found to be a separate and distinct species on the basis of several fermentative and biochemical properties. It was renamed *V. alginolyticus* in 1968. This organism was not known to be pathogenic in humans until 1973 when six isolates from tissue specimens collected in 1969 and thought to be *V. parahaemolyticus* were found to be *V. alginolyticus*. The ecological niche of this organism is probably similar to that of *V. parahaemolyticus*, and it has been isolated from seawater, fish, shrimp, crabs, oysters, and clams. The numbers of this organism in seawater are generally higher than *V. parahaemolyticus* and are associated with warm water temperatures. This organism fails to grow at temperatures less than 8°C. In the United States, clinical isolates of *V. alginolyticus* have been obtained from waters along the Atlantic, Pacific, and Gulf coasts, as well as in Hawaii and the Chesapeake Bay.

Clinically, wound and ear infections represent the majority of cases of illness due to this organism. It is the most common cause of *Vibrio*-associated wound infections in the United States. Wound infections almost always follow exposure of open wounds to seawater. Frequently, a number of different organisms may be isolated, making the pathogenic role of *V. alginolyticus* uncertain. In one study in Western Australia, 56% of infected wounds contaminated with seawater yielded *V. alginolyticus*. In a similar study in Hawaii, the organism was isolated from 11% of traumatic marine

injuries. In Florida between 1981 and 1988, 14 clinical isolates of *V. alginolyticus* were reported, of which 11 were wound isolates (79%). The incubation period is about 24 hours. Most wound infections are self-limiting and consist of mild cases of cellulitis with varying amounts of a seropurulent exudate, but more severe cases with bacteremia may be noted in immunocompromised individuals. One fatal case of bacteremia involved a 37-year-old woman who was doused in seawater after an explosion on a recreational boat. The organism was isolated from burn wounds as well as blood. The organism may produce extracellular protease and collagenase, although the role of these factors in the organism's virulence has not yet been established. Local wound care is probably sufficient for superficial wounds in immunocompetent individuals, but patients with impaired host defenses or severe or complicated infections should be treated with antibiotics. *In vitro*, *V. alginolyticus* is susceptible to tetracycline, trimethoprim/sulfamethoxazole, aminoglycosides, and chloramphenicol.

V. alginolyticus has a predisposition toward individuals with ear disorders. These infections are generally associated with swimming in seawater. Both otitis media and otitis externa have been reported. This organism has rarely been associated with gastrointestinal illness, although cases have been reported. Conjunctivitis, pneumonia, osteomyelitis, and a case of an epidural abscess due to *V. alginolyticus* have been reported. Peritonitis has been reported in an individual undergoing peritoneal dialysis who changed his peritoneal dialysis fluid bag on the beach without taking adequate precautions.

V. furnissii

V. furnissii was originally classified as biovar II of *V. fluvialis*. This species differs from *V. fluvialis* in its aerogenicity (produces gas from glucose). In 1983 it was renamed *V. furnissii* in honor of a researcher at the British Public Health Laboratories in Maidstone, England. The organism has been recovered from the marine environment. Most clinical isolates have been from Japan and other parts of Asia. Illness related to this species consists of gastroenteritis, most likely due to ingestion of raw or undercooked seafood. Occasional outbreaks of gastroenteritis have been recorded, including one episode on board a flight from Tokyo to Seattle involving 23 passengers. Symptoms included diarrhea (91%), abdominal cramps (79%), nausea (65%), and vomiting (39%). Two patients required hospitalization and one died. The organism has also been

recovered from the feces (along with *V. fluvialis*) of an infant with diarrhea and has been isolated from the stool of asymptomatic individuals. One case of bacteremia was reported in a diabetic individual who had ingested seafood in Virginia two days earlier. He was successfully treated with moxifloxacin. *V. furnissii* is capable of producing an enterotoxin similar to that described for *V. fluvialis* and *V. mimicus*.

V. metschnikovii

Previously known as enteric group 16, this organism was first described in 1888. In 1978 it was redefined on the basis of DNA homology studies. Freshwater and marine isolates of this organism have been obtained from rivers, estuaries, sewage, cockles, lobsters, oysters, clams, and a bird that died of a cholera-like illness. Human disease involving this organism was not recognized until 1978 when it was recovered from the blood of an 82-year-old diabetic female in Chicago who presented to a hospital with septicemia due to an inflamed gallbladder. She was treated successfully with a cholecystectomy and antibiotic treatment with clindamycin and tobramycin. Although the organism has been isolated from the stool of asymptomatic individuals, diarrheal illness has been described in at least one instance. The first reported case of a wound infection due to *V. metschnikovii* was reported in 2004 in a butcher who worked with farm animals in Germany. An extracellular cytolysin has been characterized, although its role in virulence is uncertain.

Vibrio cincinnatiensis

Only one case of human disease involving this organism has been reported to date. A 70-year-old man with a history of alcohol abuse developed disorientation and questionable nuchal rigidity upon admission to a hospital. A spinal tap was performed which revealed a Gram-negative bacillus; an identical isolate was obtained from his blood. Genetic and biochemical analysis revealed that this was a new *Vibrio* species which was subsequently named *V. cincinnatiensis*. The patient was treated successfully with 9 days of moxalactam. There was no history of exposure to saltwater or ingestion of seafood.

Vibrio harveyi

First isolated from a brown shark that died in captivity in 1984, this organism has been known to be a pathogen of fish and penaeid shrimp. Originally named *Vibrio carchariae* (from the Greek *carcharias*, meaning “shark”) and later renamed *V. harveyi*, this species had not been implicated as a human pathogen until a single report

described the case of a wound infection occurring in an 11-year-old girl who was attacked by a shark off the South Carolina coast. She suffered extensive trauma to her left calf, requiring plastic surgery. A swab of the leg obtained on the day of the shark bite revealed *V. harveyi*. Minimal drainage occurred from the wound, which subsequently healed well. Antimicrobial testing showed it to be susceptible to cephalothin, cefamandole, cefoxitin, gentamicin, and the combination of trimethoprim and sulfamethoxazole. Resistance was noted to ampicillin and carbenicillin; amikacin sensitivity was intermediate.

MYCOBACTERIUM MARINUM

M. marinum was first isolated by Aronson in 1926 from saltwater fish that died in a Philadelphia aquarium. In 1942, Baker and Hagan isolated this mycobacterium from freshwater platyfish in Mexico; they named it *Mycobacterium platypoecilus*. Human disease attributed to this organism was not recognized until 1951 when an epidemic of self-limiting skin granulomas involving 80 individuals occurred in a town in Sweden; 75 were children and lesions were mostly on the elbows. All five adult patients were avid swimmers, which suggested that a common swimming pool may have been involved. Tissue specimens from patients and cultures of the swimming pool yielded an atypical mycobacterium which the investigators termed *Mycobacterium balnei* (Latin meaning “of the bath”). To affirm the association between the organism and the clinical findings, the investigators inoculated themselves with the mycobacterium and were able to produce the lesions. In 1959, the previously reported mycobacterial isolates were found to represent the same species. *M. marinum* became the officially recognized name because this was the earliest term proposed by Aronson. The first association of this organism with tropical fish tanks was made in 1962. Since then, *M. marinum* has become a well-recognized cutaneous pathogen with a strong association with an aquatic environment and water-related activities. The organism has been called a leisure-time pathogen and the disease referred to as a hobby hazard. Manifestations of disease have also been referred to as fish fancier’s finger and swimming pool granuloma.

M. marinum belongs to Runyon group I (the photochromogens) of atypical *Mycobacteria* along with *Mycobacteria kansasii*, *Mycobacteria simiae*, and *Mycobacteria szulgai*. These organisms are capable of producing a yellow carotene pigment when exposed to a strong light. *M. marinum* grows optimally on Lowenstein-Jensen medium, and unlike other mycobacteria, growth

occurs at 28 to 32°C rather than 37°C. This is an important distinction which may be responsible for the fact that most *M. marinum* infections do not invade beyond the superficial cooler regions of the skin. Colonies may be visible in as early as 7 days, although it rarely may take up to 2 months. Growth also occurs on blood agar, though not on MacConkey agar. The organisms are acid fast and appear as long, slender, occasionally beaded rods, but they may appear short and compact. Biochemical reactions include a negative response to niacin, neutral red, and arylsulfatase and a positive response to Tween 80 hydrolysis in 10 days.

M. marinum is a pathogen of salt- and freshwater fish. It is widely distributed in the environment and may be isolated from contaminated water, the walls of swimming pools or aquariums, and dead or diseased fish. Chlorinated swimming pools are not considered an exposure risk. Affected fish may be cachectic and frequently show skin changes consisting of pigment changes, loss of scales, blood spots with eventual formation of ulcers, and fin and tail rot. Microscopically, miliary tubercles may be found in virtually every organ system. The disease is thought to spread among fish by the ingestion of infective material. In humans, the organism most commonly causes benign superficial cutaneous illness, although destructive tenosynovitis, osteomyelitis, septic arthritis, sclerokeratitis, and rarely, disseminated systemic illness have been reported. The organism is usually acquired through abrasions, lacerations, or punctures sustained in an aquatic environment. Commonly associated activities include swimming, boating, fishing, handling of fish and shellfish, and the keeping of tropical fish tanks. Occasional nonaquatic exposures have been documented from an asphalt school yard, school desks, an electrical installation, rose thorns, and even a laboratory. Cases have been reported along the Gulf, Pacific, and Atlantic coasts as far north as Oregon and Long Island. Inland cases also occur. An epidemic in Colorado involving 290 individuals was linked to community swimming pools.

The incubation period ranges from 2 weeks to 2 months. Superficial lesions appear which usually range in number from one to six and appear as chronic, dusky erythematous plaques or verrucoid papules or nodules in areas subject to trauma. More than 75% of reported cases involved the hand and upper extremity, usually the dominant hand. The lesions can appear as solitary granulomas or verrucous papules 1 to 2.5 cm in diameter which may ulcerate and drain purulent material. Most lesions are asymptomatic and cause only cosmetic inconvenience. Symptoms, if present, are mainly slight tenderness and discharge. Limitation of movement of

the affected extremity may be noted. Lymphadenopathy occurs rarely. Occasionally, spread in a sporotrichoid fashion may occur. Rarely, disseminated cutaneous spread develops. One such case involved a 16-month-old child who developed disseminated lesions after being bathed in a bathtub which her father had previously used to clean his fish tanks. The time lag between appearance of the lesion and its correct diagnosis ranges from a few weeks to a few years. Left untreated, 80% of the lesions may resolve completely in an average of 14 months, the longest being 45 years. Truncal cutaneous dissemination has been described in an AIDS patient who kept tropical fish as a hobby. Histologically, the lesions of *M. marinum* change their appearance according to the age of the infection. Early lesions (2 to 3 months) show nonspecific inflammatory infiltrates in the upper corium. Multinucleated giant cells and epithelioid cell granulomas generally appear at 4 to 6 months of age, and typical tuberculoid structures appear in older lesions. All stages can be observed simultaneously. Staining of tissue specimens reveals the organism in only about 10% of cases. Ten or more slides of thick smears from homogenized tissues are often required for examination after preparation with auramine-rhodamine fluorochrome stain.

Deep-tissue infections were first widely recognized in 1973. Superficial lesions may invade deeper tissues by attempted self-excision, by intralesional cortisone injections, or by incomplete surgical excision that is not combined with appropriate antibiotics. Unlike superficial lesions, these infections are more destructive and more resistant to treatment. Tendon and synovial structures of the wrist and hand are affected in most cases. Diffuse edema at the site of infection is the most common finding. A slight fullness may be palpated at the site of the affected tendon sheath. Unlike pyogenic infections, the dorsum of the hand does not swell, and a throbbing pain does not occur. Joint limitation with subsequent development of draining sinus tracts may occur. Up to one-half of the deep infections involving the hand and wrist may be associated with carpal tunnel syndrome. Systemic complaints, fever, and lymphadenopathy are usually absent. The erythrocyte sedimentation rate is usually normal and x-ray findings are nonspecific. Factors associated with a poor prognosis include persistent pain, the presence of a draining sinus tract, and previous local injection of corticosteroids.

Diagnosis relies primarily on isolation of the organism by staining or culture. A thorough history should be obtained to include such information as hobbies and recreational interests. *M. marinum* rarely grows at

temperatures used for the incubation of most other mycobacteria. If *M. marinum* is suspected, specimens should be incubated at 30°C on Lowenstein-Jensen medium (in addition to 37°C for the isolation of other mycobacteria). Specific mycobacterial skin testing has been used, although with inconsistent results. The differential diagnosis of *M. marinum* infections includes sporotrichosis, tularemia, nocardia, yaws, syphilis, leishmaniasis, common warts, coccidioidomycosis, blastomycosis, histoplasmosis, tuberculosis verrucosa cutis, sarcoidosis, gout, rheumatoid arthritis, iodine or bromine granuloma, and benign or malignant tumors.

Although usually self-limiting, superficial infections should probably be treated with antimicrobial agents. Minocycline has long been considered to be the drug of choice, but clarithromycin and doxycycline have been equally efficacious. Resistance to doxycycline has been described. *M. marinum* does not respond to many of the antimycobacterial agents. It is usually susceptible to rifampin, ethambutol, trimethoprim/sulfamethoxazole, and amikacin. Ciprofloxacin has shown good *in vitro* activity against *M. marinum*. *In vitro* susceptibility has also shown that gatifloxacin, levofloxacin, and

moxifloxacin display activities similar to ciprofloxacin. Treatment should probably be continued for at least 6 to 12 weeks. Patients should be instructed not to pick at their lesions to reduce the risk of deeper tissue involvement. Deeper tissue involvement usually requires surgical intervention, which may include excision, tenosynovectomy, synovectomy, arthrodesis, or incision and drainage of infected bone or joints. Superficial lesions should not be excised or biopsied without the addition of appropriate drug therapy to decrease the risk of deeper tissue spread.

MARINE TRAUMA

Sharks

Shark attacks have been a popular focus of many books and movies, which often depict these creatures as ruthless savage predators randomly attacking swimmers or divers along beaches or other aquatic recreational areas (Fig. 4). Between 1580 and 2015, there were 2,777 confirmed unprovoked shark attacks worldwide, of which 497 were fatal. During 2014 alone, 72 confirmed

FIGURE 4 The great white shark is one of a number of shark species implicated in shark attacks in North America. Specialized sensory systems compensate for poor color vision by allowing the shark to detect motion as well as electrical fields of its prey. Photograph © Terry Goss 2006. Licensed under CC-BY-2.5 (<https://creativecommons.org/licenses/by/2.5/>). doi:10.1128/microbiolspec.IOL5-0008-2015.f4



unprovoked attacks occurred, which was lower than the 75 unprovoked attacks reported in 2013, the lowest total since 68 in 2009. Of the 72 worldwide attacks in 2014, 52 occurred in the United States, and of these, 28 were in Florida. Of the 480 species of sharks, only 34 have been implicated in shark attacks. In the United States, the most common offenders are the great white, gray reef, blue, and mako sharks. Fatalities are mostly the result of three species: the great white, tiger, and bull sharks. Most shark attacks occur in temperate waters between 46°N latitude and 47°S latitude. Associated risk factors include murky, warm water (70°F), sewer outlets, late afternoon and early evening hours, recreational areas, deep channels or drop-offs, movement, and bright objects. The victim does not see the shark in most cases, although skin abrasions from bumping (due to the shark's skin denticles) may precede the bite. In more than 70% of reported attacks, the victim was bitten one to two times.

Sharks can range in length from 9 inches to 50 feet (the whale shark). The largest great white shark verified was 19.5 feet long, with unverified reports of great whites exceeding 21 feet. They are extremely well adapted as predators in their environment and have exceptional sensory systems that enable them to detect electrical fields and motion, which may compensate for their poor color vision. Specialized telereceptors known as the ampullae of Lorenzini are exquisitely sensitive to vibration and low-frequency sound waves, which allows these animals to detect struggling creatures. Keen olfactory and gustatory chemoreceptors allow them to sense body fluids.

The shark's jaws consist of multirowed crescent-shaped sets of rip-saw teeth which are replaced every few months. The biting force has been estimated at 18 tons per square inch. The jaw opening of some sharks may be large enough to accommodate a horse's head.

Sharks usually feed slowly and purposefully, but they occasionally become frenzied, snapping at anything in sight, usually because of an inciting event. Humans are usually mistaken for food (seals). Recent shark attacks in Northern California have been focused on surfers who had entered migratory habitats of the elephant seal, which the shark feeds on. Possible explanations for the precipitation of a shark attack include anomalous behavior by the shark, the violation of courtship patterns, or territorial invasion.

Initial management of the shark attack victim consists of basic trauma management: airway, breathing, bleeding control, and the management of hemorrhagic blood loss. The immediate threat to life is hypovolemic

shock and occasionally primary pulmonary or cardiac injury. Bleeding is controlled with compression and occasionally ligation of vessels. Wounds should be irrigated and debrided and packed open with primary closure. Tetanus toxoid, 0.5 cc intramuscularly; tetanus immune globulin, 250 to 400 units intramuscularly; and prophylactic fluoroquinolones, tetracycline, trimethoprim/sulfamethoxazole, imipenem-cilastatin, a third-generation cephalosporin, chloramphenicol, or an aminoglycoside have been indicated. Wounds are susceptible to contamination by both aerobes and anaerobes, including *Aeromonas*, *Vibrio*, and *Clostridia*. Buck and colleagues cultured the teeth of a great white shark caught off Long Island and isolated the following organisms: *V. alginolyticus*, *V. parahaemolyticus*, *V. fluvialis*, *Pseudomonas putrefaciens*, and *Staphylococcus* species. *V. harveyi* was responsible for a wound infection following a shark bite off the South Carolina coast.

Measures to be taken to avoid a confrontation with a shark include avoidance of shark-infested waters or waters near sewage outlets or deep channels (especially at night or dusk), never swimming with open wounds or in isolated waters, and storing captured fish away from divers. If confronted by a shark, the swimmer should leave the water with slow, purposeful movements, facing the shark and avoiding the urge to flee rapidly. In deeper waters, the diver should seek refuge near a wall or other object offering protection from behind or find a confined area which the shark would have difficulty accessing. The shark may desist with blows to the snout, gills, or eyes.

Barracuda

Sphyræna barracuda, or the great barracuda, is the only species implicated in human attacks. These fish may grow to 10 feet in length and weigh 100 pounds. They can be found in tropical and semitropical waters of the Atlantic (Brazil to Florida) and the Pacific (Hawaii). Barracudas involved in attacks are usually solitary, although schools have attacked. Attacks are swift and ferocious and usually occur out of confusion in turbid waters or when the fish is attracted by shiny objects. Their teeth are canine-like and sharp. The bite produces a V-shaped or straight laceration, often in rows. Treatment is analogous to shark bites.

Moray Eels

Moray eels are found in tropical, semitropical, and some temperate waters. They are fierce and muscular bottom dwellers that hide in crevices and under rocks and coral. Some species may grow up to 15 feet in length. Divers

usually encounter these creatures while probing around rocks and coral with their hands. Most smaller species flee when confronted. If provoked or cornered, an eel may inflict a serious laceration with its vice-like jaws and fang-like teeth (Fig. 5).

Occasionally, an eel does not release its grip. Attempts to disarticulate the jaw or kill the animal may be necessary. An eel's skin is leathery and may not be easily cut. Treatment is similar to that for shark bites.

Other Sources of Trauma

Needlefish are long, streamlined surface fish found in tropical waters that may attain lengths up to 2 m. They have a long sharp snout which has been reported to occasionally impale people as this fish frequently jumps out of the water. Injuries have involved the chest, abdomen, neck, and head, with one case of a fatal brain injury.

FIGURE 5 Moray eels usually flee when confronted. Fang-like teeth and a vice-like jaw allow it to inflict serious damage if confronted. Photograph © Sylke Rohrlach 2013. Licensed under CC-BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0/deed.en>). doi:10.1128/microbiolspec.IOL5-0008-2015.f5



Sea lions can occasionally be aggressive, especially mating males or females with pups. Because these animals are mammals, rabies prophylaxis should be considered.

Microbiology of Traumatic Marine Wounds

Most of the pathogenic bacteria in seawater are halophilic, heterotrophic, motile, Gram-negative rods. The genus *Vibrio* represents one such group. The potential for a wound to become infected is based on several factors including the size and extent of the wound and the colonization of the waters. Host factors tend to play a role as well, with immunosuppressed individuals more likely to develop systemic spread. One study in Hawaii of acute marine infections revealed an 11% incidence of *V. alginolyticus* wound isolates, with most of the other bacteria comprising flora typical of cutaneous infections.

Minor wound infections may do well without antibiotics, but individuals with liver disease, immunosuppressed individuals, and those with iron overload states may benefit from oral ciprofloxacin or the combination of trimethoprim and sulfamethoxazole. Antibiotics should only be initiated with specific knowledge of the circumstances and microbiology of the marine environment where the injury occurred (Table 3). Unless a wound has a classic erysipeloid appearance, penicillin, ampicillin, first-generation cephalosporins, and erythromycin should only be given if used in conjunction with another antibiotic such as a fluorquinolone, metronidazole, clindamycin, or doxycycline. Potentially serious or complicated wounds should probably be treated with intravenous antibiotics including cefoperazone, cefotaxime, ceftazidime, levofloxacin, gentamicin, chloramphenicol, and tobramycin. Fulminant infections may be treated with a combination of imipenem and cilastatin. A minor wound infection with a classic erysipeloid reaction (*Erysipelothrix rhusiopathiae*) may be treated with penicillin, erythromycin, or cephalexin. All marine injuries and wounds warrant close follow-up with extensive patient education and awareness.

VERTEBRATE ENVENOMATIONS

Stingrays

There are over 150 species of stingrays worldwide, with 11 species found in U.S. waters: 7 are found in the Atlantic Ocean and 4 are found in the Pacific Ocean. Stingrays are frequently implicated in cases of human envenomation; at least 2,000 stingray injuries

TABLE 3 Microorganisms associated with marine wound infections and recommended antimicrobial therapies^a

Microorganism	Therapy	
	First choice	Alternative
<i>Aeromonas hydrophila</i>	TMP-SMX ^b	Gentamicin, tetracycline
<i>Bacteroides fragilis</i>	Metronidazole	Clindamycin
<i>Chromobacterium violaceum</i>	Treatment should be guided by clinical presentation and <i>in vitro</i> susceptibility testing (mezlocillin and aminoglycoside have been used, according to a case report)	
<i>Citrobacter diversus</i>	Imipenem	Ciprofloxacin (or others if susceptible <i>in vitro</i>)
<i>Clostridium perfringens</i>	Penicillin G	Clindamycin, metronidazole
<i>Enterobacter cloacae</i>	Cefotaxime, ceftriaxone, or ceftizoxime + aminoglycoside	Imipenem, aminoglycoside, TMP-SMX, ciprofloxacin
<i>Erysipelothrix rhusiopathiae</i>	Penicillin G or ampicillin	Cephalosporins (e.g., cephalexin)
<i>Escherichia coli</i>	TMP-SMX	Cephalosporin; ampicillin, quinolones, aminoglycoside
<i>Mycobacterium marinum</i>	Minocycline	Rifampin + ethambutol, rifampin + TMP-SMX
<i>Providencia stuartii</i>	Cefotaxime, ceftriaxone, or ceftizoxime + aminoglycoside	Imipenem, aminoglycoside, sulfa-trimethoprim, ciprofloxacin
<i>Pseudomonas aeruginosa</i>	Aminoglycoside + antipseudomonal penicillin	Aminoglycoside + ceftazidime
<i>Salmonella enterica</i> serovar <i>Enteritidis</i>	Ceftriaxone	Ampicillin, ciprofloxacin, TMP-SMX
<i>Staphylococcus aureus</i>		
Methicillin sensitive	Semisynthetic penicillin	Vancomycin
Methicillin resistant	Vancomycin	TMP-SMX (?)
<i>Streptococcus</i> species	Penicillin G	Erythromycin, clindamycin
<i>Vibrio</i> species (see text)	Tetracycline ± aminoglycoside (if systemic spread is suspected)	TMP-SMX (possibly quinolone if susceptible <i>in vitro</i>)

^aWith rapid emergence of antimicrobial resistance, these empirical recommendations should be confirmed by *in vitro* susceptibility testing.

^bTMP-SMX, trimethoprim-sulfamethoxazole.

are recorded in the United States annually. They can be subdivided into four categories, which are (in ascending order of toxicity): the butterfly rays (gymnurid type), the eagle and bat rays (myliobatid type), the stingrays and whiprays (dasyatid type), and the round rays (urolophid type). Stingrays are found in tropical, subtropical, and warm temperate waters in shallow intertidal areas such as bays, lagoons, river mouths, and sandy areas between reefs. These marine animals, which may reach up to 50 to 60 square feet in size, are round, diamond, or kite-shaped with wide wing-like pectoral fins. They often lie on the surface bottom covered by sand, with only their eyes, spiracles, or parts of their elongated, whip-like tail exposed.

Stingrays have one to four venomous stingers arranged on the dorsum of the tail. The stingers consist of dentine spines which are tapered and retroserrated so that they can enter the skin easily but are extracted with difficulty and worsening of the laceration. Each spine is covered by an integumentary sheath which houses a venom gland in the ventrolateral groove along either side. The spine is covered by a layer of venom and mucous. The venom is a highly unstable and very heat-labile

protein consisting of at least 10 amino acids and toxic components such as serotonin, 5'-nucleotidase, and phosphodiesterase. Envenomation usually occurs when an unwary swimmer steps on the stingray while wading in shallow waters. The stingray in turn strikes the individual by lashing its tail upward and forcing its spines into the victim. The lower extremities are involved in most cases, although any part of the body is susceptible.

Initial manifestations consist of intense localized pain with soft tissue edema and bleeding. The pain may intensify and spread over a period of 30 to 90 minutes and then gradually diminish over the next 6 to 48 hours. Minor wounds resemble cellulitis, but more severe wounds may appear dusky and cyanotic with progression to rapid hemorrhage and necrosis of fat and muscle. Secondary infection is common.

The venom affects the cardiovascular system and can cause peripheral vasoconstriction or dilatation and cardiac arrhythmias. Respiratory depression (via the medullary centers) and convulsions may also occur. Other symptoms include nausea, vomiting, generalized edema (with truncal wounds), limb paralysis, and hypotension.

Initial treatment should consist of irrigation with cool water or saline to produce local vasoconstriction and debridement with exploration to remove any sheath contents left in the wound. X rays should be obtained to identify any missed fragments. The wound should be soaked in hot water (113°F) for 30 to 90 minutes. The benefit of this probably has to do with the thermolabile nature of the protein. Narcotics and infiltration with 1 to 2% lidocaine without epinephrine may provide pain control. Regional-nerve anesthesia with 0.5% bupivacaine may be necessary. The wound should be packed open for primary closure or sutured loosely. Antibiotic prophylaxis is recommended. Steroids and antihistamines are without documented efficacy.

Scorpion Fish

Scorpion fish can be found in shallow reef waters of the Gulf of Mexico, the Florida Keys, and the coastlines of California and Hawaii. Members of this group are responsible for about 300 envenomations annually in the United States. Several hundred species are known to exist which can be divided into three genera based on the structure of the venom glands: *Synanceja* (stonefish), *Scorpaena* (sculpin, scorpion fish, and bullrout), and *Pterois* (zebra fish, lionfish, and butterfly cod). Scorpion fish often hide under rocks, in coral crevices, or buried in the mud blending in excellently with their surroundings due to their colorful and ornate camouflage. Stonefish are unattractive bottom dwellers, which are regarded as the most lethal of these poisonous marine denizens. The potency of their venom is comparable to cobra venom. A small number of domestic envenomations occur in tropical fish hobbyists who carelessly handle illegally obtained scorpion fish. The venom organs consist of 12 to 13 dorsal, 2 pelvic, and 3 anal spines. The ornate pectoral spines are not venomous. Paired glands exist at the base of the anterolateral spines, and venom flows along grooves in these spines. Approximately 5 to 10 mg of venom may be found in each paired gland. When threatened, these fish erect their dorsal spines, flare out their armed gill covers, and protrude their pectoral and anal fins. Venom is injected in a manner analogous to a stingray envenomation. The potency of the venom varies according to the species. The pharmacology of scorpion fish venom is poorly understood. Wounds inflicted by lionfish are mild compared to scorpion fish, and stonefish wounds are the most severe.

Intense pain occurs immediately after the injury and radiates up the extremity. If untreated, the pain peaks in 60 to 90 minutes, often lasting 6 to 12 hours or perhaps days. Stonefish envenomation may produce pain severe

enough to precipitate delusions. The wounds initially appear ischemic and cyanotic with surrounding areas of erythema, edema, and warmth. Progressive cellulitis and local induration may follow, occasionally with vesicle formation. Tissue necrosis with sloughing may occur within 48 hours. Wounds may take months to heal, with occasional soft-tissue fibrous defects or cutaneous granulomata persisting. Secondary infection or deep abscesses may occur. Systemic effects include paresthesias, skin rash, nausea, vomiting, arthralgias, fever, diarrhea, delirium, seizures, abdominal pain, hypertension, arrhythmias, limb paralysis, congestive heart failure, hypotension, and death. In the case of stonefish envenomation, dyspnea and circulatory collapse may occur within 1 hour, with death ensuing within 6 hours.

Initial treatment of scorpion fish injuries is analogous to stingray injuries, with proper irrigation and debridement to ensure removal of any persistent source of venom. Inadequate debridement may predispose to ulceration with tissue extension. Immersion in hot water (113°F) for 30 to 90 minutes may alleviate the pain that is possibly secondary to toxin inactivation; longer durations may be needed for persistent pain. Wounds should be packed open for primary closure or sutured loosely to allow drainage. Antibiotic therapy is recommended for deep puncture wounds of the hand or foot because of the high incidence of tissue complications. Recommended antibiotics include third-generation cephalosporins, trimethoprim/sulfamethoxazole, chloramphenicol, tetracycline, aminoglycosides, or imipenem/cilastatin. Severe systemic reactions due to *Synanceja* species (and less rarely due to other scorpion fish) necessitate the use of intravenous stonefish equine antivenin, which is supplied in 2-ml ampules, with 1 ml capable of neutralizing 10 mg of dried venom. Antivenins are produced in Australia and India and may be acquired in the United States from the Health Services Department; Sea World in San Diego, California; Sea World in Aurora, Ohio; and the Steinhart Aquarium in San Francisco, California. Physicians may alternatively locate antivenin by contacting an accredited regional poison center.

Sea Snakes

Sea snakes are probably the most abundant reptile in the world. There are 52 known species, and all are venomous. At least seven species have been implicated in fatal envenomations. The sea snake is a marine-adapted serpent belonging to the family *Hydrophidae* and is a close relative of the cobra and krait. They are widely distributed in tropical and subtropical waters along the

coasts of the Indian and Pacific Oceans and the Gulf of California. There are no sea snakes in the Atlantic Ocean or Caribbean Sea. These reptiles have a flat tail, which assists them in propulsion. They can travel with ease in both a forward and reverse direction. Some species may attain lengths of up to 10 feet. Sea snakes can remain submerged in the water for hours, using an air retention system to control buoyancy. Although usually docile, they may attack when provoked, especially during the mating season. Sea snakes have two to four maxillary fangs, each of which is associated with a pair of venom glands. The fangs of most species are too short to penetrate a diver's wet suit. Many envenomations are avoided by the fact that the fangs are easily dislodged. The venom is more potent than terrestrial snake venom, and its toxicity is derived primarily from potent neurotoxins. Neurotoxins may exert their effects through the inhibition of cholinesterase or by blocking presynaptic or postsynaptic impulses. Most sea snake venom contains only postsynaptic neurotoxin, which strongly binds to the acetylcholine receptor at the neuromuscular junction. The venom may contain other toxins including hyaluronidase, phosphodiesterase, phospholipase A, and proteases. Phospholipase A and other myotoxins may cause muscle necrosis with myoglobin release. These myotoxic, hemolytic, neurotoxic and vasoactive components constitute a serious medical emergency if envenomation occurs.

The bites are extremely small and characterized by multiple pinhead, hypodermic-like puncture wounds numbering from 1 to 20. Initially, the pain is minimal without any local reaction. Neurologic symptoms occur within 2 to 3 hours in most cases and always within 8 hours of the bite. If no symptoms develop within 8 hours, significant envenomation did not occur. Typical signs and symptoms include painful muscle movement, myoglobinuria, euphoria, trismus, malaise, anxiety, ascending paralysis, slurred speech, dysphagia, ptosis, and ophthalmoplegia. Death is rare and is usually due to more severe reactions including myonecrosis; respiratory insufficiency; bulbar paralysis; and hepatic, cardiac, or renal failure.

Diagnosis is based on several factors such as location (sea snake bites only occur in the water), absence of pain (initial pain is unusual), the typical appearance of the fang bites (usually one to four), identification of the snake (it should be caught if possible), and the development of characteristic symptoms within 8 hours (occasionally as early as 5 minutes).

Treatment is similar to that for most terrestrial snake bites. Local suction without incision can be performed

if used immediately with a plunger device. Incision and drainage are of no value if delayed for several minutes after the bite, and cryotherapy is contraindicated. The affected limb should be immobilized with application of a proximal venous and lymphatic occlusive band or use of the pressure immobilization technique (a cloth pad is pressed directly over the wound by a circumferential elastic wrap at a pressure of ≤ 70 mm Hg) to minimize systemic toxicity. Antivenom is indicated for shock, respiratory failure, hyperkalemia, leukocytosis, severe myalgias or pain with movement of the extremities, trismus, myoglobinuria, elevated creatinine kinase levels, and altered level of consciousness. Antivenom is effective if used within 24 to 36 hours and should be used quickly. The minimum initial dose of sea-snake antivenom is 1 to 3 vials; up to 10 vials may be required. Polyvalent equine antivenom to *Enhydrina schistosa* (beaked sea snake) and *Notechis scutatus* (terrestrial tiger snake) will neutralize the bites of most sea snakes. Anaphylaxis and serum sickness may occur, and intradermal sensitivity testing with 0.02 ml of a 1:10 antivenom dilution should be done. *N. scutatus* antivenom is a second-choice alternative. In the United States antivenom may be obtained from the sources listed in the section on scorpion fish. If antivenom is not available, dialysis should be considered because sea snake neurotoxin is of low enough molecular weight to be dialyzable.

Weeverfish

These fish, also referred to as sea cat, sea dragon, ad-derpike, and stang, are found in the temperate waters of the Atlantic Ocean, Mediterranean Sea, and European coastal waters. Weeverfish are one of the most venomous fish found in these waters. They are small fish (<50 cm) that can be found buried in the mud or soft bottom with only their head exposed. They produce venom located in glands associated with dorsal and opercular dentine spines, which are capable of piercing a leather boot. Weeverfish are generally docile, yet if provoked, they may become extremely aggressive and strike. Victims are usually professional fisherman or wading swimmers.

The venom is a heat-labile protein substance which also contains 5-hydroxytryptamine, histamine, epinephrine, and norepinephrine. An intense burning pain is felt immediately following the sting. The pain often spreads throughout the entire limb and generally peaks in intensity within the first hour and usually subsides within 24 hours, although it may persist for days. The initial wound is pale and edematous and becomes erythematous, warm, and ecchymotic with increasing edema. It may take months to heal. Headache, delirium,

fever, dyspnea, diaphoresis, nausea, vomiting, seizures, hypotension, and cardiac arrhythmias may occur.

Treatment is analogous to stingray envenomations. The pain is often poorly controlled even with the liberal use of narcotics. An antivenom may be available soon. Weeverfish should never be handled when alive; they may survive for hours out of water.

Catfish

More than 1,000 species of freshwater and saltwater catfish exist. These fish are named for the sensory barbels surrounding the mouth. Catfish have long thin venom glands covered by an integumentary sheath located on lateral ridges along the dorsal and pectoral spines. Catfish may lock these fins in an extended position when excited, thereby inflicting a dramatic sting if mishandled. Fragments of tissue and spine may remain in the wound. The venom contains dermonecrotic, vasoconstrictive, and other bioactive agents. Immediate severe pain, throbbing, and weakness occur. The pain usually radiates locally and proximally and generally resolves in 30 to 60 minutes, occasionally lasting 48 hours. Nausea, vomiting, respiratory distress, and mild hypotension may occur. Muscle fasciculations and local muscle spasms are common. The wound site may appear pale and bleed longer than expected. Cyanosis at the puncture site may occur, followed by local tissue necrosis or skin sloughing. Swelling of the extremity, a serous discharge, slow wound healing, and local lymphadenopathy may persist for weeks to months. Secondary infections are common and gangrene has been reported. Envenomation by the oriental catfish (*Plotosus lineatus*) may produce more marked systemic symptoms.

Treatment is analogous to stingray envenomations, although the venom is not as potent. X rays should be obtained because the radiopaque spines are frequently left in place. The wound should be thoroughly irrigated and debrided. Irrigation with permanganate, bicarbonate, acetic acid, or papain have not been shown to be of benefit. Tourniquets have no value. Updating of tetanus immunization, as in other marine puncture wounds is essential. Antibiotics should be administered if the wound is dirty or complicated. Local wound care alone is probably adequate for properly treated clean wounds in healthy individuals.

INVERTEBRATE ENVENOMATIONS

Coelenterates

The phylum *Cnidaria* (formerly *Coelenterata*) is a massive group of marine invertebrates consisting of more

than 10,000 species, of which at least 100 are hazardous to humans. The classes are *Scyphozoa* (true jelly fish), *Hydrozoa* (fire coral, Portuguese man-of-war, Pacific blue bottle), and *Anthozoa* (sea anemones, soft coral). These species may contain millions of nematocysts (also known as stinging capsules or nettle cells), which are microscopic toxin-containing organs capable of injecting venom into prey when triggered. Nematocysts are found along the outside of long streaming tentacles and near the mouth of the animal. Each nematocyst consists of a minute capsule (cnidoblast) that contains a spiral-coiled thread with a barbed end. Specialized triggers (cnidocil) on the capsule are capable of forcefully ejecting the venomous barbs at pressures up to 2 to 5 lb/in², enough to penetrate a surgical glove. This produces a severe "sting." The size of the barb and the potency of the venom varies between the different species. The cnidocil may be triggered by abrasion or freshwater contact. A single man-of-war envenomation may involve several hundred thousand nematocysts.

The human reactions to jellyfish toxin may be classified as systemic, local, chronic, or fatal. Local reactions include persistent or recurrent eruptions, eruptions distant to the site of envenomation, exaggerated local angioedema, papular urticaria, and contact dermatitis. Chronic reactions include keloid formation, pigmentation, fat atrophy, mononeuritis, autonomic nerve paralysis, gangrene, vascular spasm, and ataxia. Systemic reactions may include muscle cramps, nausea, vomiting, tachycardia, abdominal pain, and hypertension. Death may occur through cardiotoxic, central respiratory, or renal mechanisms. Anaphylaxis has been reported.

The potency and composition of the toxin varies among species. After injection of large doses of toxin into human skin, a perivascular mononuclear cell infiltrate appears within the dermis. A cell-mediated and humoral immune response then occurs. Lymphokines and prostaglandins account for some of the cutaneous manifestations of jellyfish envenomation. Serotonin, histamine, or histamine-releasing agents are responsible for much of the burning pain and urticaria. High-molecular-weight toxins may inhibit nerve activity through altered ionic permeability. These protein and tetramine fractions have direct and indirect effects on the autonomic nervous system and end organs, particularly the vascular, cardiac and central nervous systems. A destabilizing effect on the cell membrane appears to be mediated by the calcium channel. Verapamil may block this effect.

Fire corals (*Millepora*) are not true corals. They are branching bottom dwellers which may attain heights

of 1 to 2 meters. They develop upright, clavate, blade-like, or branching calcareous growths that form encrustations over rocks, shellfish, other coral, and man-made objects. The white to yellow-green lime carbonate exoskeleton may become razor sharp. The nematocyst-bearing tentacles protrude from numerous minute surface gastropores. These stings account for the majority of coelenterate envenomations. An intense burning pain, with central radiation and reactive regional lymphadenopathy follows envenomation.

The Portuguese man-of-war (*Physalia physalis*) and the Pacific blue bottle (*Physalis utriculus*) are not true jellyfish; they are colonial hydroids. These open-sea surface creatures are widely distributed in the Atlantic (*P. physalis*) and Pacific (*P. utriculus*) Oceans and are most commonly found in the semitropical Atlantic Ocean. They are composed of a large nitrogen- and carbon monoxide-filled sail which may achieve lengths up to 30 cm across. Numerous nematocyst-laden tentacles (up to 750,000 nematocysts per tentacle) stream downward from the sail, attaining lengths of up to 30 meters (*P. physalis*). The animal is carried along by wind and ocean currents. Fish and other objects may become entangled in the tentacles, which contract rhythmically in search of prey. Broken-off tentacle fragments may retain their potency for months.

Chironex fleckeri (box jellyfish or sea wasp) is the most venomous sea creature known. Death may occur within 1 minute of envenomation, and the overall mortality has been reported to be as high as 20%, although this number may be inflated as evidenced by more recent studies. The sting from this animal is described as an excruciatingly painful burning sensation much like being branded with a red-hot branding iron. These creatures are found in the protected waters off the northern Australian coast. There have been 63 confirmed deaths to date due to stings from these creatures. A less lethal variety is found in the waters of the Chesapeake Bay. Up to 10 ml of venom may be injected during a sting. They are small creatures (2 to 10 cm in diameter) capable of attaining speeds up to 2 knots in steady winds with swift currents. Most of the deaths have been attributed to the larger animals. Factors determining the severity of envenomation include the length and width of the wheal, the length of the contact, the thickness of the skin, the percentage of nematocysts discharged, and the venom load, which is related to recent feeding. A sheep-derived antivenom is available and is given intravenously or intramuscularly (20,000 units, or one vial, intravenously over five minutes or three vials intramuscularly). Verapamil is contraindicated.

Sea anenomes are sessile, multicolored, flower-like anthozoans which may attain diameters of up to one-half meter. They have finger-like projections containing numerous modified nematocysts referred to as sporocysts. They are usually encountered by skin divers and waders.

The Irukundji syndrome may follow stings by the *Cubozoa* class jellyfish (*Carukia barnesi*). A small but transient sting, often like a mosquito bite, initiates this reaction. A severe, boring pain may then develop hours later in the abdomen or sacrum, with gradual spread to the thighs and chest, where the sensation is described as cramping. Severe, excruciating myalgia persists over the next 24 to 48 hours. Other symptoms include tremor, anxiety, piloerection, hyperpnea, headache, nausea, vomiting, sweating, restlessness, tachycardia, blood-streaked sputum, and oliguria. A rapidly developing pulmonary edema leading to acute respiratory failure may develop. This syndrome, similar to excess catecholamine release, is assumed to represent a toxic reaction to the venom of the small jellyfish.

The person who has endured a coelenterate sting should immediately have tentacles and large fragments removed gently with a forceps or gentle scraping. Double gloving is recommended. Once the wound is cleaned, shaving cream should be applied and the area can be shaved gently with a razor or credit card. Rubbing the area should be avoided because this may trigger nematocyst release. Hot water immersion in the range of 104 to 113°F (40 to 45°C) has been shown to be effective for stings of the Hawaiian box jellyfish and the Portuguese man-of-war and may be effective for those of other jellyfish as well. Vinegar has been shown to be effective, but this has generated some controversy because it may trigger venom release. Isopropyl alcohol (40 to 70%), once considered a mainstay of treatment, may also further venom release. Other detoxicants of possible benefit include dilute ammonium hydroxide, sodium bicarbonate, olive oil, sugar, urine, and papain (unseasoned meat tenderizer). Fresh water should never be applied to the area because this may trigger further nematocyst release. The area should be soaked for at least 30 minutes or until the pain disappears.

Anaphylaxis should always be anticipated. After decontamination, corticosteroids or topical anesthetics can be used. Antibiotics are usually not needed, but large open lesions should be cleaned daily and covered with a thin layer of nonsensitizing antiseptic ointment. Tetanus immunization should be current. Wounds should be checked by a physician for signs of infection 3 and 7 days after the injury.

Echinoderms

Sea urchins and starfish are included in this group of marine animals. The venom contains many toxic substances including steroid glycosides, serotonin, and acetylcholine-like substances. Certain sea urchins may produce potent neurotoxins.

Sea urchins are globular or flattened animals with a hard shell enclosing their vital organs. Regularly arranged spines and a triple-jawed seizing organ (pedicellariae) cover this shell. Spines may be venomous or nonvenomous. The pedicellariae are dispersed among the spines and may grab hold with their pincer-like jaws. They also contain venom glands that release toxic material when they contract. The venom inflicts intense burning stings, which may progress to muscular paralysis, respiratory distress, and occasionally death if numerous spines are involved. Hot water may provide relief. The pedicellariae and embedded spines should be removed with care because they are easily fractured. Residual spines may form granulomas.

Starfish are simple, free-living, stellate echinoderms covered with simple thorny spines of calcium carbonate crystals held erect by muscle tissue. Glandular tissue interspersed throughout or located beneath the integument produces a slimy, venomous substance that causes a contact dermatitis. Envenomation occurs when the victim contacts the thorny spines, some of which may grow to 6 cm in length. Envenomation can rarely induce systemic symptoms including paresthesias, vomiting, and muscular paralysis. The dermatitis can be treated with hot water and topical calamine with 0.5% menthol.

Mollusks

Cone shells are potentially lethal gastropods that possess a sophisticated venom apparatus. At least 18 to 400 species have been implicated in human fatalities. They are nocturnal feeders located in the Indo-Pacific area. A set of minute harpoon-like radular teeth may contain venom, which is injected from an extensible proboscis. The venom interferes with neuromuscular transmission in a manner analogous to curare. Initial symptoms include local ischemia, cyanosis, and numbness. More severe envenomations may induce paresthesias, and generalized muscular paralysis with respiratory failure. Other symptoms include dysphagia, aphonia, weakness, diplopia, blurred vision, cerebral edema, disseminated intravascular coagulation, coma, and cardiovascular collapse. Death may occur in as little as 2 hours. Therapy is largely supportive. Hot water immersions may alleviate some of the pain.

Octopuses are cephalopods found in the warm waters of the intertidal zone. The Australian southern blue-ringed (*Hapalochlaena maculosa*) and greater blue-ringed (*Hapalochlaena lunulatus*) octopuses have been implicated in human fatalities. The blue-ringed octopus is covered with poorly visible blue rings, which become iridescent peacock blue when the animal is angered. Parrot-like and powerful chitinous jaws are capable of penetrating through the dermis of the skin and into the muscle tissue. Venom is injected into the victim, who is usually a naive swimmer playing with what seems to be a harmless creature. The toxin blocks nerve conduction most likely by altering sodium conductance. Myocardial and respiratory depression are observed in animal models. Following the initial bite, an intense burning or throbbing with central radiation occurs. Within 30 minutes marked local erythema, swelling, pruritus, and pain may occur. Severe envenomations include nausea, vomiting, paresthesias, blurred vision, aphonia, dysphagia, ataxia, myoclonus, flaccid paralysis, hypotension, and respiratory failure. Treatment is supportive with wide excision of the involved area down to deep fascia. Closure may be primary or by a full-thickness skin graft. There is no effective antivenin available.

Dermatitis

Most cases of dermatitis in swimmers and beachgoers are the result of contact with cercariae, nonhuman schistosomes. In 1991, an outbreak of cercarial dermatitis was noted among 37 students who frequented a Delaware beach. In 1999, 63 confirmed cases of cercarial dermatitis were reported among 450 families who had been swimming near a snail-infested beach near Quebec City. Cercarial dermatitis, or swimmer's itch, is a cutaneous inflammation due to penetration of the skin by cercariae, which are the free-living larval stages of bird schistosomes. Hosts include migratory waterbirds, including shorebirds, ducks, and geese. Adult worms are carried in the bloodstream and produce eggs that are passed in the feces. Once exposed to the water, the eggs hatch to produce miracidium which infect mollusks. The parasite develops in the snail to produce cercariae, which penetrate the skin of birds to complete the cycle. Humans are accidental hosts. The cercariae are able to penetrate the skin but do not develop further.

This dermatologic entity has a worldwide distribution. Symptoms include reddening and itching of exposed skin in the water or immediately after emerging and delayed onset of intensely pruritic raised papules

which may form vesicles days later. Previous exposure may elicit a more severe response upon re-exposure. Treatment consists of antihistamines and topical anti-pruritic medications.

Cutaneous larva migrans is a dermatitis caused by invasion of the skin by larval nematodes. Most infections are due to the filariform larvae of the dog and cat hookworm, *Ancylostoma braziliense*, although several other larval nematodes may cause the disease including *Ancylostoma caninum*, *Ancylostoma duodenale*, and *Necator americanus*. Adult *A. braziliense* inhabit the intestines of dogs and cats. They produce eggs which pass in the feces to hatch in the soil in 1 to 2 days. Within a week, they become infective filariform larvae. Humans are inadvertent hosts when the larvae penetrate the skin, although they usually do not penetrate further than the epidermis. The disease has a worldwide distribution and is much more common in tropical and semitropical areas. People who frequent beaches are at increased risk (hence the synonym sandworm).

Symptoms develop within a few hours of penetration into the skin. An itching red papule develops which may form a serpiginous track. The surrounding tissues become edematous and acutely inflamed. The tracks may become encrusted and secondarily infected. The pruritus may become extremely intense. Untreated, the larvae may persist in the skin for months. Löfflers syndrome as a result of extensive cutaneous larva migrans has been reported. Although most cases are self-limited, treatment consists of a single dose of ivermectin. Children may be treated with albendazole.

Seabather's eruption is caused by skin penetration of the larva of certain *Cnidaria*. Thimble jellyfish (*Edwardsiella lineata*) and the lined sea anemone (*Linuche unguiculata*) have been implicated, but other *Cnidaria* may be implicated. Also known as sea lice, pika-pika, sea critters, and ocean itch, these are salt-water organisms which can cause an intensely pruritic rash as a result of hypersensitivity to toxins discharged by the larval nematocysts. Larvae may be trapped in the material of swimsuits, swimming caps, swimming fins, cuffs of wet suits, and T-shirts. Usually, the rash appears at the site of the material. In the United States, most cases occur along the coasts of the Atlantic Ocean and Caribbean Sea.

Symptoms of seabather's eruption may develop in minutes to 12 hours after swimming and occur when clothing begins to dry and nematocysts become physically disrupted. Freshwater may also discharge nematocysts, which explains why some cases don't appear until showering after exposure. The rash appears as inflammatory papules or blisters of different sizes and shapes

that can become extremely erythematous and pruritic. Headaches, nausea, vomiting, myalgias, malaise, and fatigue may accompany the rash. Urethritis and conjunctivitis have been reported. Treatment is largely supportive and usually consists of oral antihistamines and topical steroids. Prevention in endemic areas includes prompt removal of wet material and showering with warm seawater if available. Clothing should be washed with laundry detergent.

SWIMMER'S EAR

Swimmer's ear, or acute otitis externa, is the most common medical problem faced by swimmers. It begins initially not as an infection, but as an eczema of the ear canal caused by retention of water in the ear following bathing, showering, or swimming. By the time medical attention is sought, the ear is usually secondarily infected. The normal external auditory canal is sterile bacteriologically in up to 30% of the population, with the remainder harboring mixed flora including *Staphylococcus albus*, *Staphylococcus epidermidis*, diphtheroids, and to a lesser extent, *Staphylococcus aureus* and *Streptococcus viridans*. In external otitis media, cultures usually reveal mixed flora, with Gram-negative bacteria predominating in up to three quarters of affected individuals under the age of 21. *Pseudomonas* species, the most common offending organisms, are isolated in up to one half of the cases. Other species include *Proteus vulgaris*, *E. coli*, *S. aureus* (including methicillin-resistant strains), *S. epidermidis*, streptococci, diphtheroids, *Enterobacter aerogenes*, *K. pneumoniae*, and *Citrobacter*. Approximately 40% of infected ears yield fungal isolates, with *Aspergillus* species representing the majority.

The pathogenesis of swimmer's ear is multifactorial, with cerumen playing a major role. Cerumen imparts an acid reaction to the external canal, lowering its pH to 5 and thereby inhibiting bacterial and fungal growth. In addition, its lipid content provides a protective surface to the squamous epithelium and pilosebaceous elements of the canal. Cerumen therefore provides a chemical and mechanical barrier to infection. Excessive moisture in the external canal, which can occur during swimming, bathing, or excessive sweating, can lead to mechanical disruption of this barrier with subsequent desquamation and maceration. In addition, the decrease in cerumen results in an increase of the pH to 7, which allows bacterial species to proliferate. As inflammation occurs, a purulent exudate forms which mixes with the dry skin, providing a wet environment suitable for bacterial growth.

The earliest symptom is usually itching, which often leads to manipulation. Trauma from cotton swabs or mechanical objects inserted into the canal can exacerbate the condition. A purulent discharge develops accompanied by progressive tenderness and pain. Hearing loss may result from canal skin edema and accumulation of debris. The diagnosis can be confirmed by manipulation of the tragus and pinna, which elicits a severe painful reaction. The pain, as well as the accumulation of debris, may hamper attempts at otoscopic visualization. Regional lymphadenopathy and cellulitis of the external auricle can occasionally occur. Itching and a feeling of fullness may be the only symptoms of fungal otitis externa. In these cases, hyphae may be visualized on microscopic examination, or in the case of *Aspergillus niger*, a grayish membrane may be noted in the external canal.

Initial treatment should begin with the recognition and elimination of specific precipitating factors, such as swimming or manipulation of the ear, until the disorder is corrected. Pain control may be necessary during the first 24 to 48 hours of treatment. Severe cases may require narcotic analgesics or even hospitalization for intramuscular analgesics and intravenous antibiotics. Thorough irrigation and cleaning is essential to remove the purulent debris and allow penetration by topical antibiotics. Various cleansing solutions have been used including 3% acetic acid, 70% alcohol, 3% boric acid—70% alcohol solution, and Burow's solution. Cleansing and irrigation may be required on a daily basis. In cases where the debris is too thick, an expanding cellulose wick may be implanted within the canal to allow antibiotic penetration.

Topical antibiotics are usually adequate in all but the most severe cases, which may require systemic agents. The most widely prescribed otic solutions contain the antibiotics neomycin and polymyxin, which are effective against most of the usual pathogens. Neomycin is effective against *Proteus* and *Staphylococcus* species; polymyxin is usually effective against *Pseudomonas*. Fluoroquinolones, particularly ciprofloxacin and ofloxacin, have excellent coverage of the most common pathogens. The combination of ciprofloxacin and dexamethasone may be superior to the combination of neomycin, polymyxin, and hydrocortisone in terms of decreasing swelling, pain, and edema. Chloromycetin otic drops are available for less commonly encountered anaerobic infections. In the event of otitis externa due to *A. niger*, local application of amphotericin B, oxytetracycline and polymyxin, iodochlorhydroxyquin, and nystatin may be effective. Candidal infections may

be treated with topical nystatin. Methicillin-resistant *S. aureus* otitis externa has been successfully treated with aural toilet along with fucidic acid and betamethasone drops if the organism is gentamicin resistant or aminoglycoside-steroid containing drops if it is gentamicin sensitive. Most otic preparations also contain an acidifying agent(s) and a topical steroid to reduce the inflammation. Wicks should be placed in severe cases and replaced every 1 to 3 days if needed.

Individuals prone to otitis externa who are frequently exposed to water may benefit from any of several commercially available earplugs and other devices, which prevent water from entering the canal. A recent comparison of seven ear protectors used by a group of swimmers found the most effective plugs to be cotton wool coated in paraffin jelly. Prevention may be further enhanced by the use of acidifying drops and/or drying eardrops before and after swimming. Domeboro, Swim Ear, Aqua Ear, Ear Magic, and VoSol Otic drops are several such agents. Alternatively, individuals may make their own preparation by mixing white vinegar with 70% alcohol. Controversy had existed with regard to swimming with tympanostomy tubes in place. While some experts advocated the use of protective earplugs or avoidance of water immersion, others had recommended pre- and postexposure antibiotic otic drops. The first evidence-based guidelines to specifically address tympanostomy tubes in children were published in 2013. The guidelines recommend that clinicians no longer encourage routine prophylactic water precautions (use of earplugs, headbands; avoidance of swimming or water sports) in children with tympanostomy tubes.

VIRAL HEPATITIS AND OTHER VIRUSES ASSOCIATED WITH WATER AND SHELLFISH

Marine samples of water have been estimated to yield between 5×10^6 and 15×10^6 total viruses per milliliter. Even though the majority of these viruses are non-pathogenic in humans, viral agents associated with hepatitis and gastroenteritis are known to be present in and capable of producing disease through contaminated water. Samples of water taken off the Texas Gulf Coast have revealed the presence of enteroviruses such as coxsackievirus, echovirus, poliovirus, and hepatitis A. These enteroviruses were detected in over 40% of waters deemed safe for recreational use by fecal coliform standards. Enteroviruses were detected in 35% of waters approved for shellfish harvesting. Enteroviruses have been reported to survive 2 to 130 days in seawater in

laboratory studies and possibly up to 18 months. Temperature seems to be the most important factor, with many enteroviruses able to survive for months at temperatures below 10°C. Marine sediments also protect against virus inactivation, probably by reducing the rate of thermal inactivation.

Although controversy exists, swimming areas have been suggested to be involved in the spread of some common viral illnesses. In Wisconsin, a statistically significant increase was noted in the use of public beaches by children with documented enteroviral illness compared to children without enteroviral illness. Several other reports have similarly suggested increased enteroviral infections in swimmers using public swimming areas. A survey of oysters contaminated by enteroviruses in Japan found nearly identical viruses in sick children in the immediate area. The authors suggested that water and oyster contamination ultimately depend on the prevalence of enteric viral infections in local inhabitants. In addition to more common and less severe enteroviral illnesses, an outbreak of hepatitis A involving 20 campers in Louisiana was associated with swimming in a contaminated swimming pool.

Shellfish harvested from contaminated waters have been implicated in numerous outbreaks of food-borne illness. They spread disease by virtue of their ability to filter large amounts of water, retain filtered products in the gills and alimentary tract, and accumulate them in the liver. Shellfish have the ability to concentrate hepatitis A virus up to 15 times the level in the immediate water. Commercially harvested waters are monitored for contamination through the use of bacterial counts. Although this method effectively decreases the incidence of illness related to contaminated shellfish, there is not a reliable correlation between bacterial counts and the presence of pathogenic viruses. Several methods of detecting viruses or viral particles have been investigated, although the use of any one method of testing may not be sufficient to provide accurate measurements. Furthermore, viral testing has proven to be costly and time-consuming. Control measures have included the regulation of drainage effluent into monitored areas, supervision of drainage from commercial fishing boats, frequent testing of water and shellfish from harvest areas for fecal coliforms and other bacteria, closing of contaminated waters, and removal of contaminated shellfish from markets. Depuration measures in which bacteria and viruses are removed by bathing in rapidly circulating salt water treated continuously by ultraviolet radiation for up to 72 hours have decreased the incidence of food-borne outbreaks, although this process is not

infallible. Outbreaks of viral diseases have been attributed to depurated shellfish. Prevention of illness additionally relies on public education, particularly in regard to proper cooking of shellfish. It has been shown that it takes up to 6 minutes of steaming for the internal temperature of clams to reach that of the immediate surroundings. Clams will open their shells in less than one minute. Hepatitis A virus has been shown to be inactivated by heating at 85°C for 1 minute and partially inactivated at 60°C for 60 minutes. Recent evidence suggests that microwaving may be beneficial in inactivating hepatitis A, but further studies are needed.

Hepatitis A Virus

Hepatitis A was first linked to the ingestion of raw shellfish in Sweden in 1956; 629 oyster-associated cases were documented. The first cases of hepatitis A linked to raw shellfish consumption in the United States occurred in 1961; both clams and oysters were involved. Hepatitis A caused less than 7% of the 224 recorded outbreaks of waterborne disease in the United States between 1971 and 1978; it caused none of the outbreaks recorded in 1981. During 1983 to 1988, the incidence of reported cases of hepatitis A increased from 9.2 to 10.9 cases per 100,000 population, which represented the first increase in hepatitis A in more than a decade. In 1988, 26,600 cases of hepatitis A were reported in the United States, 7.3% of which were associated with food-borne or waterborne outbreaks. In August of 1988, 61 people developed hepatitis A after ingesting oysters illegally harvested from coastal waters off Bay County. That same year, the largest documented epidemic to date occurred in Shanghai, China, in which more than 288,000 people developed hepatitis A after ingesting raw or improperly cooked clams. In 2012, there were 1,562 cases of hepatitis A reported in the United States, which was an increase of 11% from 2011. This included 468 hospitalizations and 6 deaths. There were 13 food and water outbreaks of hepatitis A infection reported that year.

Shellfish implicated in cases of hepatitis A include oysters, clams, cockles, and mussels. In addition, the illness has been traced to contaminated lettuce, raspberries, ice-slush beverages, and community water sources. Sporadic cases also occur, and the disease is probably underreported. Restaurant-related outbreaks due to poor personal hygiene by infected food preparers have also been noted. In many cases, a source of the outbreak is never determined.

The incubation period of hepatitis A ranges from 3 to 6 weeks. Presentation consists of a prodrome of fatigue,

malaise, anorexia, nausea, and right upper quadrant discomfort. Dark urine subsequently develops with liver enlargement and biochemical evidence of hepatitis. Icteric disease may be prolonged in adults, lasting 4 to 6 weeks; fulminant hepatic failure is rare, and chronic disease or a carrier state does not develop. Secondary cases in families may be seen. Attack rates vary, ranging from 10 to 97%, with higher rates associated with increasing dose and age. A vaccine is currently available and is indicated for all children between 12 and 23 months of age, anybody over the age of 1 traveling to or working in areas with high rates of hepatitis A endemicity, children and adolescents 2 through 18 years of age living in one of the 20 states with higher rates of hepatitis A, homosexual men, people who use street drugs, people with chronic liver disease, people who are treated with clotting factor concentrates, people who work with hepatitis A-infected primates or in research labs, and people living in a household that is planning to adopt a child from an area of high hepatitis A endemicity. Vaccination should be considered for children or adolescents living in a community where a hepatitis A outbreak is occurring, unvaccinated people exposed to hepatitis A, and anyone over the age of 1 year who desires protection.

Norovirus

Noroviruses are the most common infectious cause of acute gastroenteritis following shellfish ingestion. The Norwalk virus, which is the only species of the genera *Norovirus*, is named after the town in Ohio in which an outbreak of acute gastroenteritis occurred in 1968. It was not until 1972 that this virus was first identified by immune electron microscopy of bacteria-free stool isolates from the 1968 outbreak. Norovirus can be divided into five genogroups based on sequence homology; GI, GII, and GIV include the human pathogens. Within each genogroup are multiple genotypes. Genotype II consists of 17 classified genotypes (GII.1 to XII.17), but GII.4 is the only genotype associated with global epidemics, as well as higher death rates and hospitalizations than any of the other genotypes. In 2006, two GII.4 variants, designated 2006a and 2006b, were implicated as the predominant causes of norovirus-associated gastroenteritis across Europe in 2006. New GII.4 strains emerge every 2 to 3 years, replacing the previously predominant GII.4 strains. Emergence of these new norovirus strains often leads to increased outbreak activity. In March 2012, a new GII.4 norovirus, named GII.4 Sydney, was identified in Australia and has emerged as a major cause of acute gastroenteritis

outbreaks in multiple countries. In the United Kingdom, an early onset of the 2012 winter norovirus season was associated with the emergence of GII.4 Sydney as the dominant strain implicated in outbreaks. In the United States, GII.4 Sydney has spread rapidly nationwide and is presently the predominant cause of norovirus outbreaks in the United States. During September to December of 2012, GII.4 Sydney was responsible for 141 of the 266 norovirus outbreaks in the United States.

Norovirus was identified as the cause of 42% of the 74 outbreaks of acute, nonbacterial gastroenteritis investigated by the CDC between 1976 and 1980. Six additional outbreaks involving more than 820 people were noted between 1981 and 1983. Drinking water and recreational water used for swimming were the vehicles of transmission in most of these earlier outbreaks. Contamination by human sewage was the most likely explanation, although this was never proven.

Shellfish-associated noroviral illness was first recognized in Australia in 1978 when 2,000 people developed acute gastroenteritis following raw oyster ingestion. Immune electron microscopy of 93 stool specimens revealed the 27-nanometer viral particle in 39% of those examined. In 1982, 103 well-documented outbreaks of oyster- and clam-associated illness involving 1,017 people were attributed to norovirus in New York state. Seroconversion to norovirus antibody or a positive response for IgM antibody to the virus (or both) were noted in five of seven outbreaks; clams and oysters revealed the norovirus antibody by radioimmunoassay in four of six specimens examined, including clams and oysters from two of the outbreaks. Coliform testing of the waters from which the shellfish were harvested were well within acceptable limits. In 2002, 21 outbreaks of acute gastroenteritis due to norovirus on 17 cruise ships demonstrated how easily norovirus can be transmitted from person to person in a closed environment. Cruise ships are now required to log and report any passengers or crew who report gastrointestinal illness or buy anti-diarrheal medication. The CDC usually conducts an investigation when 3.0% or more of the ships passengers or crew report illness.

Incubation times have ranged from 24 to 48 hours in most recorded outbreaks, with duration of illness ranging from 2 to 60 hours (mean 12 to 60 hours). Transmission is spread person to person and indirectly through contaminated food and water. Airborne transmission may occur through aerosolization following vomiting or toilet flushing. Symptoms have consisted of nausea, vomiting, diarrhea, abdominal cramps, and headache, with vomiting being more prominent than

diarrhea in children and diarrhea being more common than vomiting in adults. Fever, myalgias, chills, and upper respiratory complaints such as a sore throat, cough, and runny nose were also noted on occasion. The disease is rarely severe enough to require hospitalization, and there have been few fatalities.

The antibody response is not entirely understood. Antibody prevalence appears to peak within the first five decades of life, reaching prevalence in the United States of 50%. Children tend to acquire antibody earlier in less developed countries. Prior GII infection offers some protection, which may be of importance for vaccine development. Immunity from illness is incomplete and temporary and may last 6 months, but is usually gone by two years.

The mechanism by which the norovirus causes diarrhea is not known, but it may involve apoptosis of enterocytes. Related strains, serotypes, and isolates, often referred to as Norwalk-like, include the Hawaii, Snow Mountain, Mexico, Desert Shield, Southampton, Lordsdale, and Wilkinson virus, named after the places they were first described. Norwalk, Desert Shield, and Southampton viruses belong to genotype 1(GI). Illnesses caused by these norovirus agents are indistinguishable from each other in terms of presentation. These other agents are antigenically distinct, and antibodies are not cross-protective.

Hepatitis E Virus

Hepatitis E virus is the only known cause of enterically transmitted non-A, non-B hepatitis. Transmission occurs through the fecal-oral route. Hepatitis E is largely waterborne and has been responsible for epidemics of infectious hepatitis in parts of Asia, Indonesia, Africa, and Mexico, where it is widespread. It is estimated to cause around 20 million infections a year, with 3 million acute cases and 57,000 deaths. Previously known as enterically transmitted non-A, non-B hepatitis, illness due to hepatitis E was first documented in New Delhi, India, in 1955 when 29,000 cases of icteric hepatitis were identified following fecal contamination of the city's water supply. A similar epidemic occurred in 1975 in Ahmedabad, India, when the city's water supply became fecally contaminated as well. Both epidemics were thought to be due to hepatitis A virus until retrospectively analyzed paired serum samples from documented cases revealed in 1980 that neither hepatitis A nor B were responsible for the epidemics. It now appears that in developing countries, more than 50% of acute viral hepatitis appears to be caused by agents other than hepatitis A or B viruses. Numerous epidemics of

hepatitis E have been reported in Pakistan, Bangladesh, Nepal, Burma, Borneo, Algeria, Somalia, Sudan, Ivory Coast, Mexico, China, Egypt, Ethiopia, and the former Soviet Union. In the United Kingdom, cases of hepatitis E have been linked to domestic pigs after being detected in 10% of retail pork sausages. Subsequent studies identified the virus in 49% of pigs in Scotland. Imported cases have been reported in the United States. One case acquired in San Jose, California, has been described. There have been 12 cases of hepatitis E reported in the United States to date without any link to contaminated food or travel to hepatitis E –hyperendemic countries.

The incubation period appears to be 2 to 9 weeks, with an average of 6 weeks. Attack rates have been variable, with the highest rates noted in young to middle-aged adults (15 to 40 years old). The clinical presentation is that of a self-limiting disease resembling infection due to hepatitis A virus. Chronic liver disease or persistent viremia has not been reported. Mortality has been reported to be between 0.5 to 3.0% in the general population. Pregnant women in the third trimester of pregnancy have an unusually high mortality of up to 20% for unknown reasons. In contrast, pregnant women infected with hepatitis A infection have a mortality ranging from 3 to 8%. Hepatitis E may cause a chronic infection in immunocompromised individuals, particularly in those who have undergone solid organ transplantation. In some of these transplant recipients, progression to cirrhosis and fulminant liver failure has occurred.

The diagnosis of infection due to hepatitis E virus has primarily been based on the identification of the virus in stool samples by immune electron microscopy. Hepatitis E is a spherical, nonenveloped virus 32 to 34 nanometers in size with a genome consisting of a single strand of polyadenylated RNA. Identification of viral particles in clinical samples has been inconsistent, possibly due to proteolytic degradation or susceptibility to freezing and pelleting. Cynomolgus monkeys, and more recently owl monkeys, have been shown to be promising animal models of the serologic response of infection by the hepatitis E virus. Radioimmunoassay or enzyme-linked immunoassay for hepatitis E antigen using acute-phase or convalescent-phase IgM or IgG have previously met with little success, possibly due to the lability of the virus. More recently, an enzyme-linked immunosorbent assay based on clonal recombinant hepatitis E antigen has been developed to detect hepatitis E IgG and IgM antibodies. A vaccine is currently available outside of the United States for individuals in endemic areas.

PRACTICAL TIPS

- When visiting coastal regions, be aware of environmental conditions and any current or recent restrictions that may be or have been in place such as red tide warnings or sewage contamination.
- Recognize the signs and symptoms of seafood-related illnesses and be sure to obtain seafood only from a reputable source.
- Swim in marked areas and never dive alone.
- Enjoy the marine aquatic environment from a distance and do not disturb its natural features; leave it exactly as you find it so that others may admire its beauty as well.
- Plan ahead by identifying the nearest medical facility and keep a first aid kit handy; make sure to share any history of recent seafood ingestion or marine aquatic travel with any medical caregivers if you are injured or ill.

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